

WorkshopPlus: Azure AI Platform and Services

Azure AI Foundry

Agenda

- ❑ Azure AI Foundry – Introduction
- ❑ Models
- ❑ Azure AI Services
- ❑ Deployment Options
- ❑ Complete AI ToolChain
- ❑ Agents
- ❑ Evaluation
- ❑ AI Innovation across Landscapes
- ❑ Use Cases
- ❑ What is new in Azure AI Foundry



Objectives

After completing this Learning, you will be able to:

1	Understand.... ✓ Describe capabilities of Azure AI Foundry Unified AI Platform for Developers and IT	
2	Understand.... ✓ Use and leverage Azure AI Foundry's Comprehensive Model Catalog and Flexibility	_____
3	Understand.... ✓ How Azure AI Foundry serves as a complete toolchain for AI Application Development and enables secure data integration, customization and experimentation	_____
4	Understand.... ✓ How Azure AI foundry serves as a facilitator for rapid development and deployment of AI Agents that can automate business processes, integrate with enterprise data, and operate with enterprise-grade security and compliance.	_____
5	Understand.... ✓ Built-in evaluation, Governance and Responsible AI	_____
6	✓ Support for multiple programming languages and integration with AI Fabric	

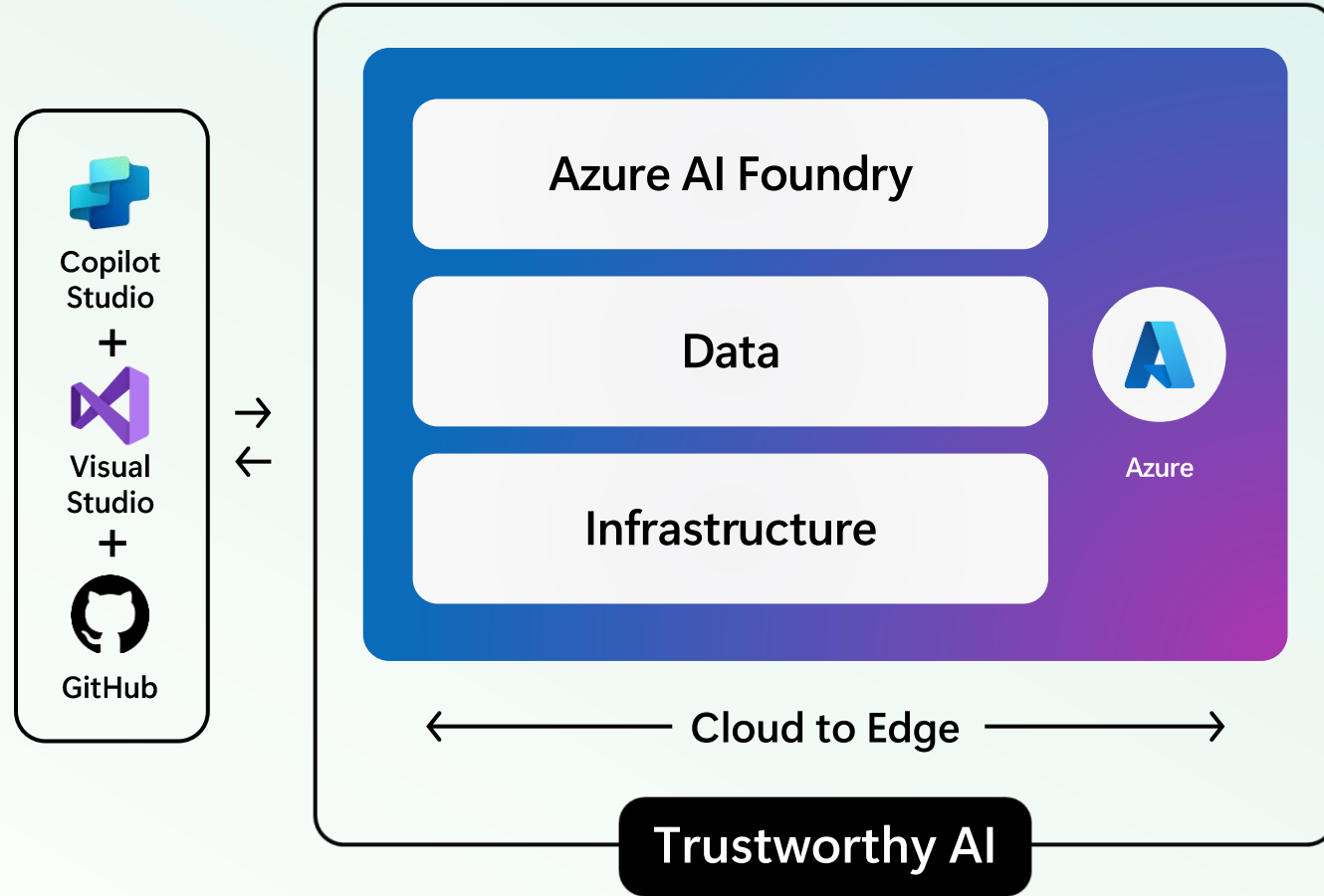


Conditions and terms of use

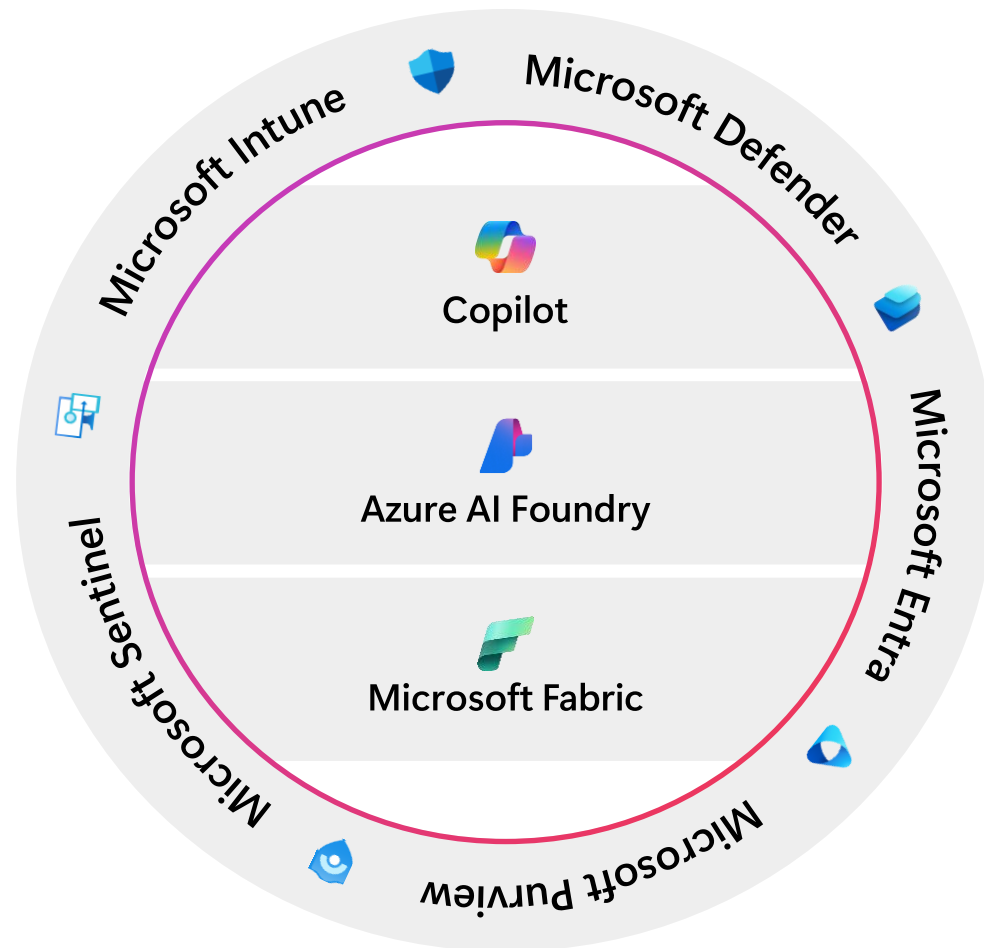
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Copilot & AI stack



AI transformation with Microsoft



World's most loved developer tools



Visual Studio



GitHub



Copilot Studio

Microsoft runs on Azure AI Foundry

Microsoft
Copilot

Microsoft 365
Copilot

Microsoft Security
Copilot

GitHub
Copilot

DAX
Copilot



Azure AI Foundry



Copilot Studio



Visual Studio



GitHub



Azure AI
Foundry SDK



Model Catalog

Foundational models

Open-source models

Task models

Industry models



Azure
OpenAI Service



Azure
AI Search



Azure AI
Agent Service



Azure AI
Content Safety



Azure Machine
Learning

Evaluations

Customization

Governance

Monitoring

Observability



Azure AI Foundry agent ecosystem

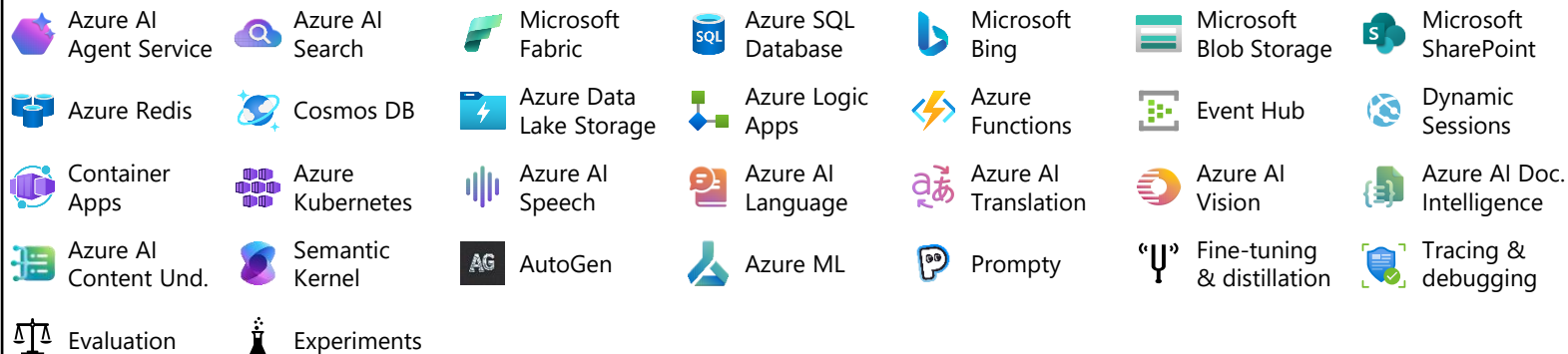


Design with the best models

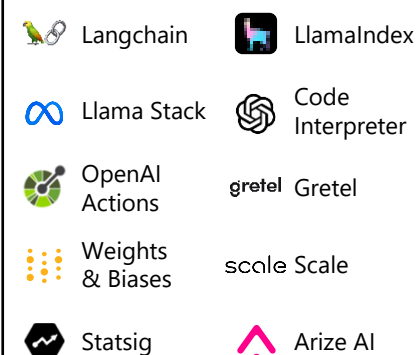


Customize with a comprehensive agent toolchain

1P offerings



3P offerings



Manage performance in production



Safeguard with Trustworthy AI



Azure AI Foundry SDK

Unified toolchain for AI application development

Access our leading models
through a single interface

Easily integrate Azure AI
capabilities into your applications

Develop faster with a
simplified coding experience

[AI.Azure.com](https://ai.azure.com)

Azure AI Foundry SDK for streamlined development



Create an
AI project to
connect services
and data



Use a common
inferencing API
for our most
popular models



Use agents
to unlock another
level of intelligence

Automatically take actions,
including self-correct

Integrate your existing data
and services (Bing Search,
Azure AI Search, Microsoft
Sharepoint, Microsoft Fabric)

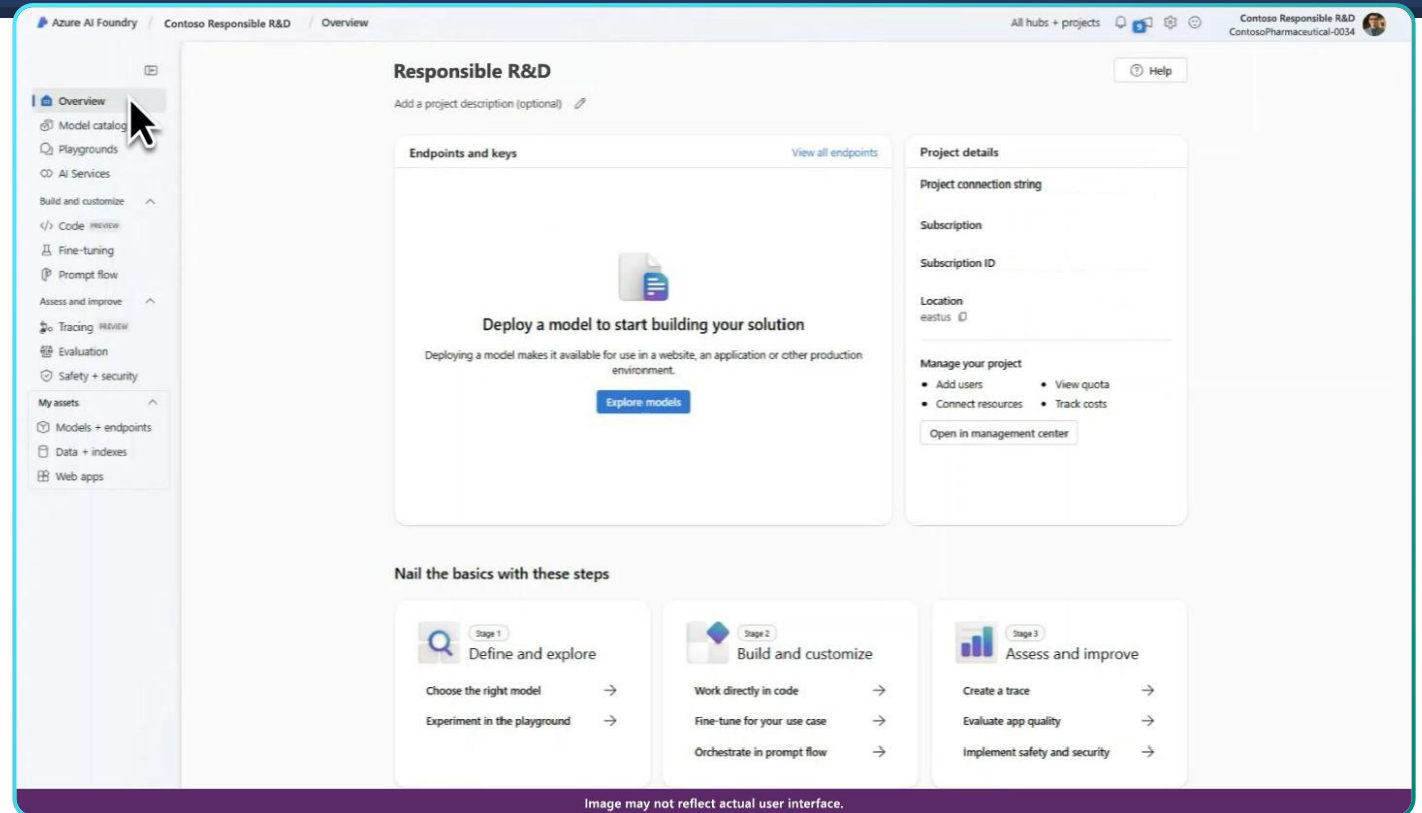


Deploy apps
to Azure using
pre-built
templates

Azure AI Foundry SDK

The Azure AI Foundry SDK is a comprehensive toolkit for developers, offering pre-built modules and resources to integrate AI functionalities seamlessly into applications.

- Access our most popular models through a single interface
- Easily integrate Azure AI Foundry capabilities into your application with a single project client
- Unlock another level of intelligence with Azure AI Agent Service
- Evaluate your apps locally, in the cloud, and in production using state-of-the-art safety and quality evaluators
- Review project logs with integrated tracing
- Access Azure AI App Templates beyond SDKs, to provision resources for popular agentic scenarios, hosted in web, container, function app, and more



The Azure AI Foundry SDK provides a local developer experience that reduces the complexity of using multiple resources together in code when building AI apps and agents.

Top use cases for Azure AI Foundry



Build your own agent



Chat with your data



Analyze and summarize documents



Generate code and documentation



Deliver personalized recommendations



Improve information discovery



Generate engaging content

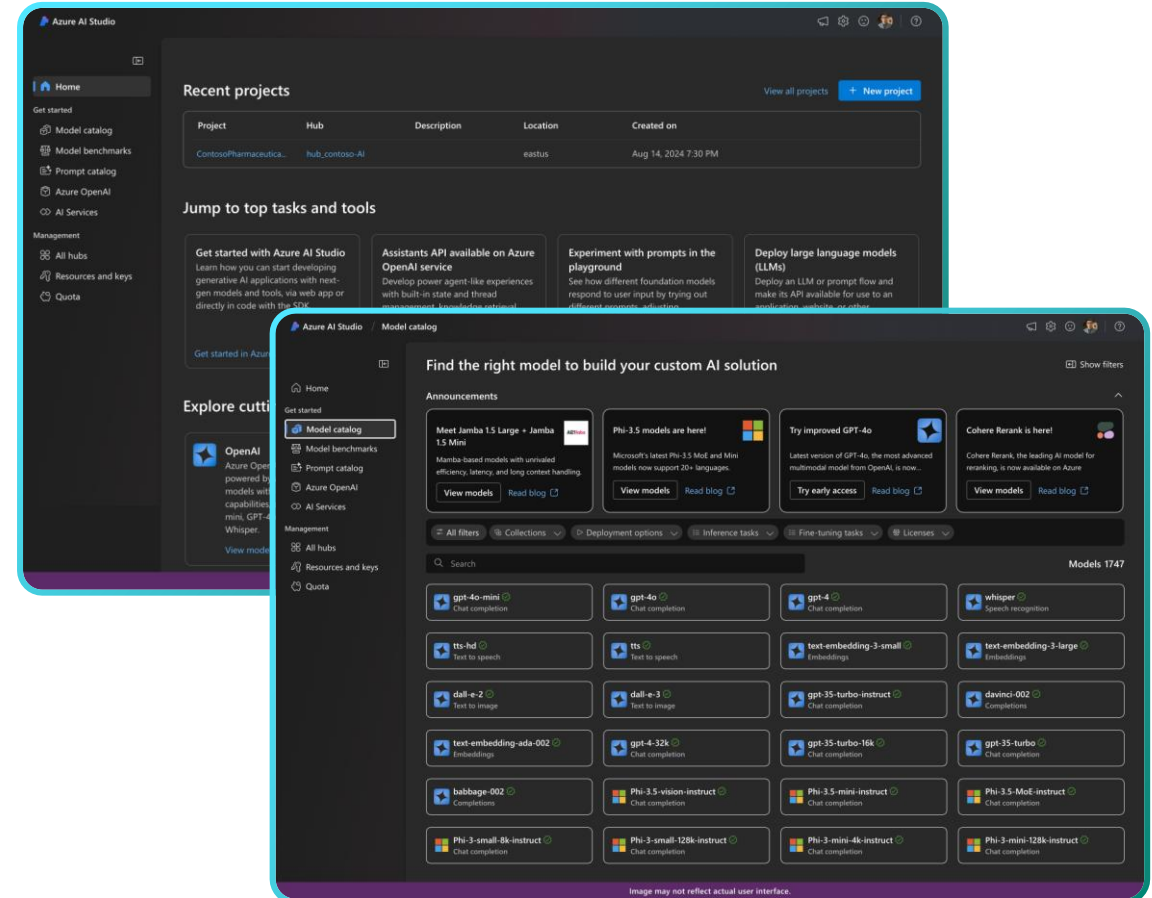


Process transactions and detect fraud



Improve forecasting

Accessibility to empower developers of all abilities to thrive in the age of AI



Azure AI Foundry was designed and built with feedback from developers with disabilities

Design with the best models

AI innovation requires flexibility



Limited
flexibility



Complex
integration



Resource
constraints



Inefficient
performance



Scalability
issues

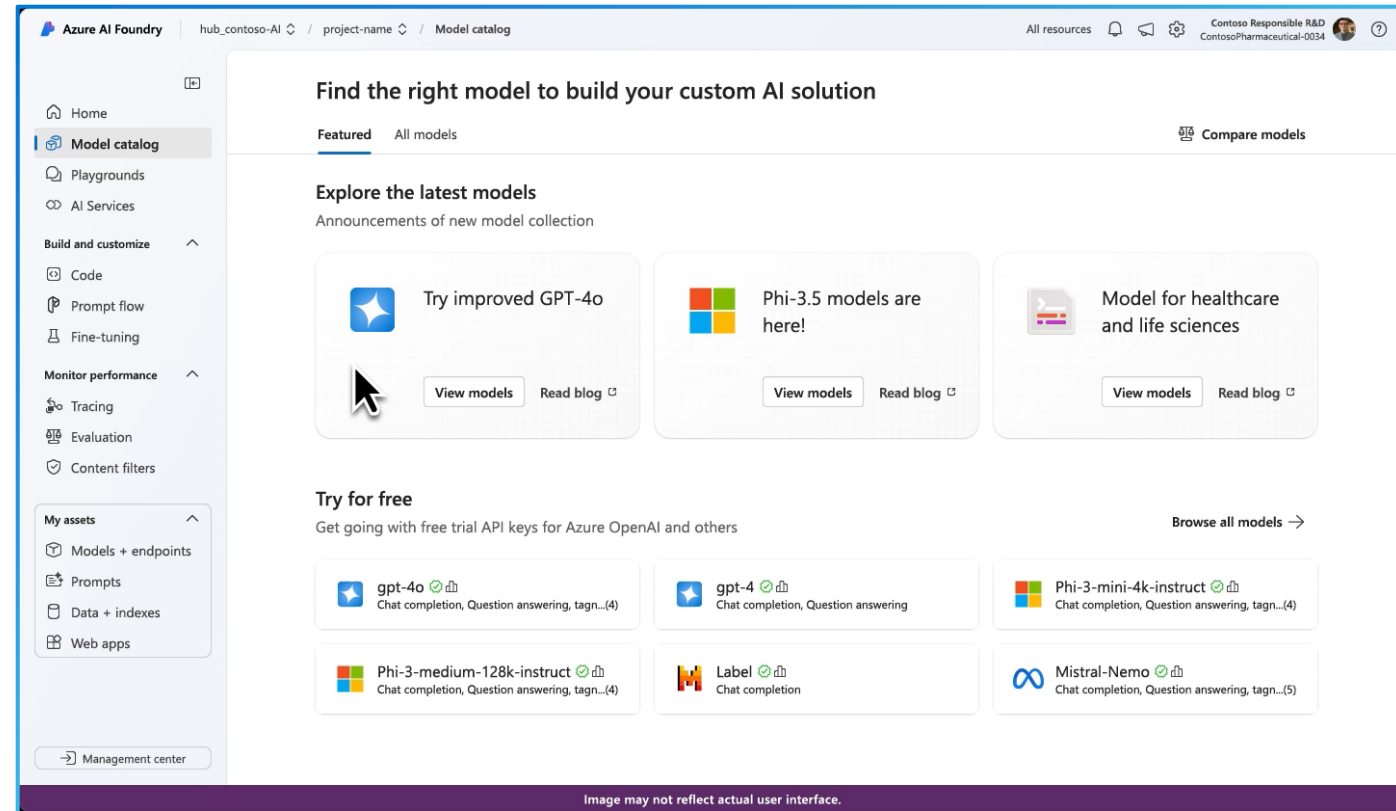
Design with the best models



Discover models for your
use case

Design intelligent apps with
industry-leading task models

Deploy serverless models



Design with the best models

Identify the best AI services and models for your use case

Discover models for your use case

Choose from the latest cutting-edge foundation, open, task, and industry-specific models. Benchmark models for your use case by performance and key parameters. Easily change models without coding.

Build intelligent apps with industry-leading task models*

Develop multimodal, multi-lingual AI with out-of-the-box and customizable APIs and models.

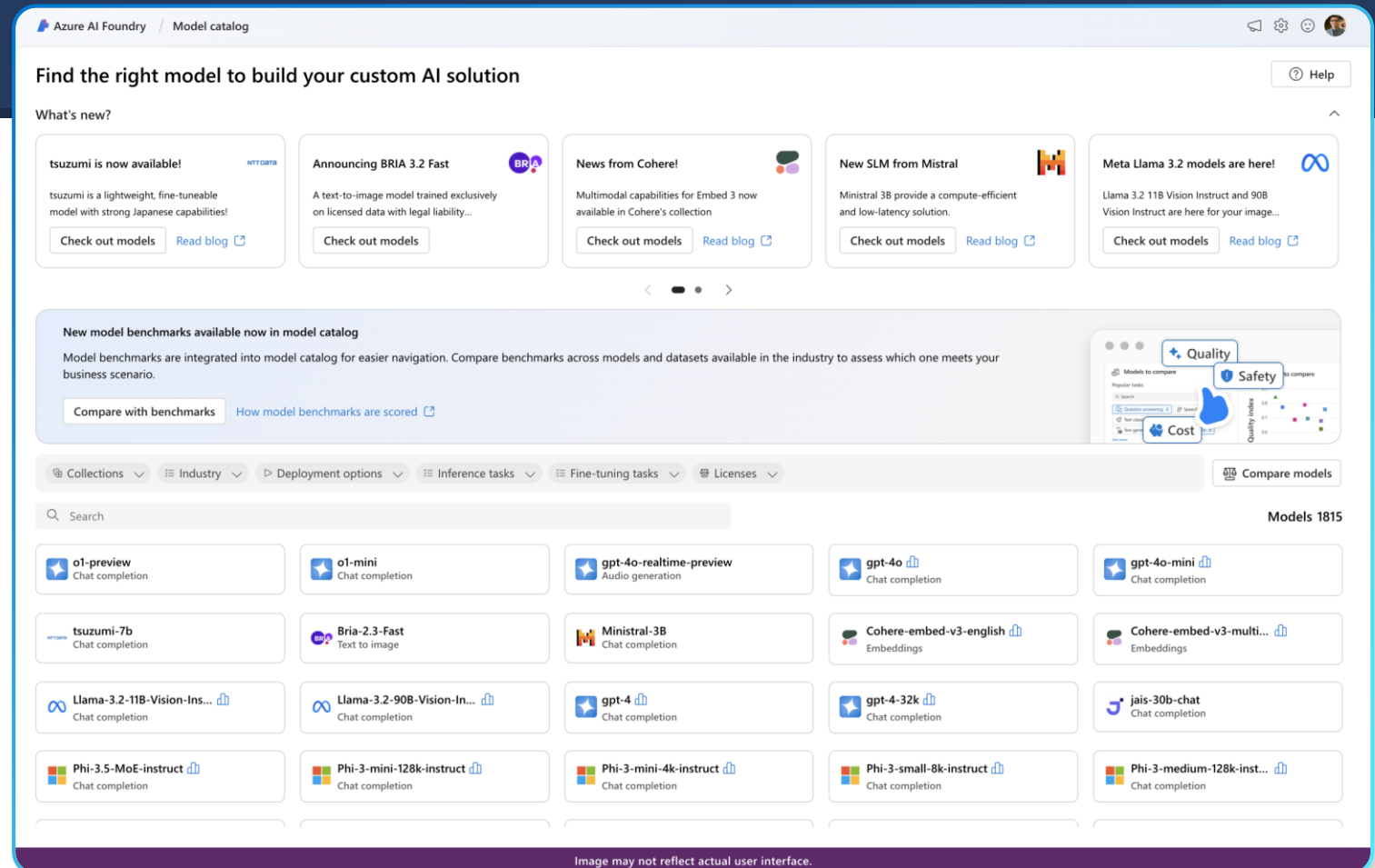
Deploy serverless models

Get started quickly with ready-to-use APIs, without the need to provision GPUs.

Model catalog

The model catalog enables discovery of large and small language models packaged for out-of-the-box or custom usage in Azure AI Foundry.

- Models from OpenAI, Hugging Face, and Meta, including Llama-3.1-405B and 3.2 series, Azure OpenAI o1 and gpt-4o series.
- Filter model catalog by collection, industry, tasks, or model name to find the model that best suits your needs.
- Compare models by Benchmarks, now integrated into Model Catalog: benchmarks across models and public datasets to find the model that best meets your business scenario.
- Redesigned model card that highlights key information about the model.
- Leverage ready-to-use, curated fine-tuning pipelines, code-based inferencing, and evaluation of the model



Easily find the foundation model that best suits your needs while improving business outcomes, reducing operational expenses, and enhancing competitive advantage.

Azure AI model breadth

Azure OpenAI Service



O3 mini – for reasoning
o1-preview – for reasoning
o1-mini – for reasoning
GPT-4o
GPT-4o mini
GPT-4-Turbo with Vision, GPT-4, GPT-3.5
DALL-E
Embeddings
Whisper, Text to speech

Phi models



Phi-3-mini
Phi-3-small
Phi-3-medium
Phi-3-vision
Phi-3.5-mini
Phi-3.5-vision
Phi-3.5-MoE
Phi-4
Phi-4-mini
Phi-4-omni

Meta



Meta-Llama-3-70B/70B-Instruct
Meta-Llama-3-8B/8B-Instruct
Meta-Llama-3.1-70B/70B-Instruct
Meta-Llama-3.1-8B/8B-Instruct
Meta-Llama-3.1-405B-Instruct
CodeLlama-7b/13b/34b/70b-Instruct/Python-hf

Mistral AI



Mistral Large
Mistral 7B
Mixtral 8x7B – Mixture of Experts
Ministral
Codestral

Cohere



Cohere R+
Cohere R
Cohere Rerank
Command and Embed
Embed v3-Multilingual
Embed v3-English

Hugging Face



Falcon/TII
Stable Diffusion/Stability AI
Dolly/Databricks
CLIP/OpenAI

Databricks



Databricks/dbx-base
Databricks/dbx-instruct

NVIDIA



Nemotron-3-8B-4k
Nemotron-3-8B-Chat-SFT/RLHF/SteerLM
Nemotron-3-8B-QA

Snowflake



Snowflake/arctic-base
Snowflake/arctic-instruct

Others

AI21 Jamba 1.5 Large + Jamba 1.5 mini
Core42 JAIS
deci (Deci AI)
Nixtla TimeGEN-1
SDAIA (Arabic Model)
Bria AI
NTT Data Tsuzumi
DeepSeek-R1

Find the best model for every application

Avoid model lock-in:

- Flexible model selection
- Curated for enterprise
- Explore, compare and swap models quickly
- Test before deploying

1800+ frontier, task and open models
in Azure AI Foundry model catalog



OpenAI
Model Family
(available day 1)



Phi SLM
Model Family



Mistral AI
Model Family



Meta Llama 2
Model Family



Jais G42
Model Family



Snowflake
Model Family



Cohere
Model Family



Databricks
Model Family



Hugging Face
Collection



NVIDIA
Model Family



Deci AI
Model Family



Nixtla
Time-Gen1 Model



NTT Data
Model



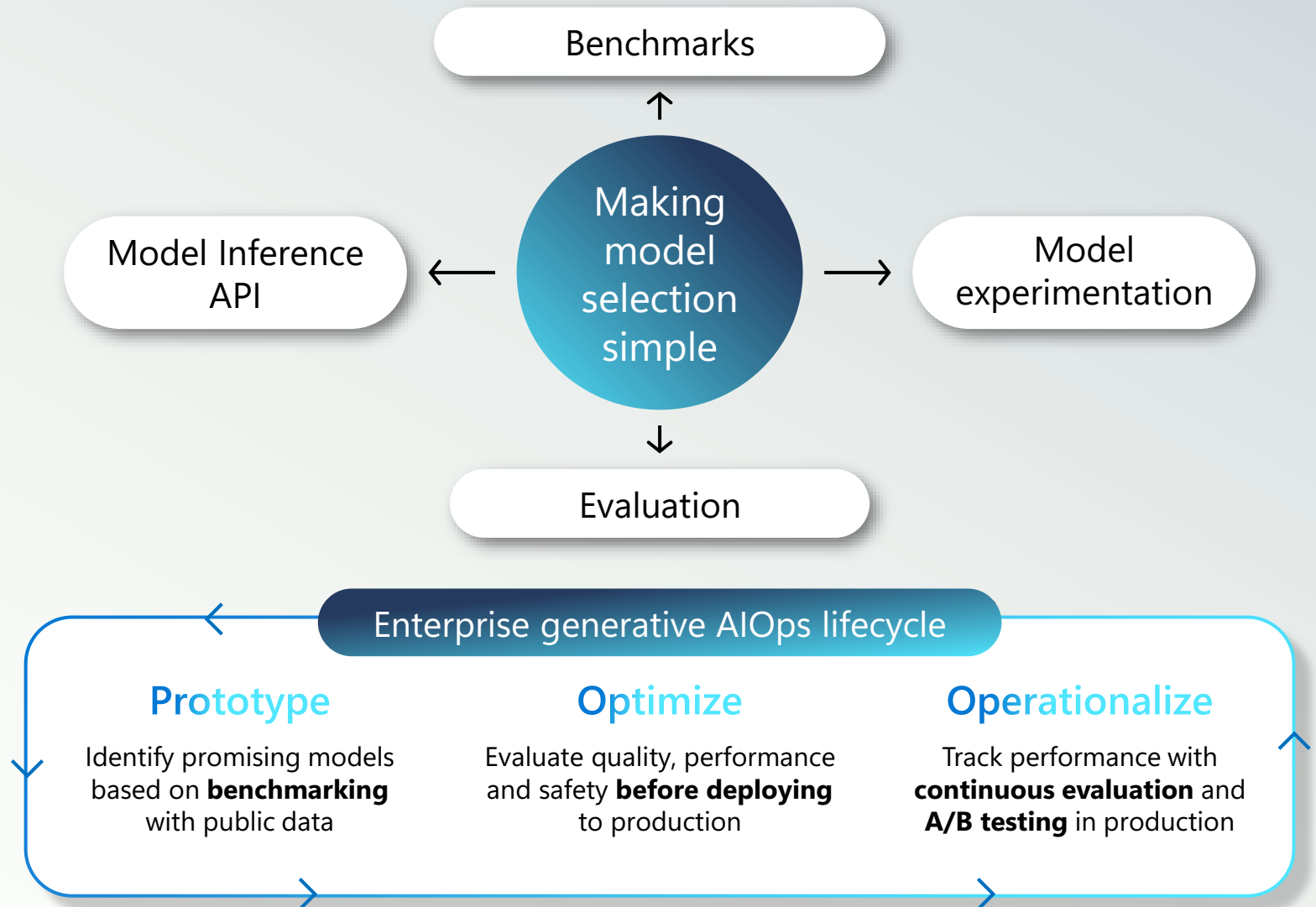
Industry models:



Select your best models with support through every stage of your AI projects

The Azure AI Foundry difference:

- Unified API for model swaps
- Built-in, full-lifecycle tooling
- Versatile deployment options

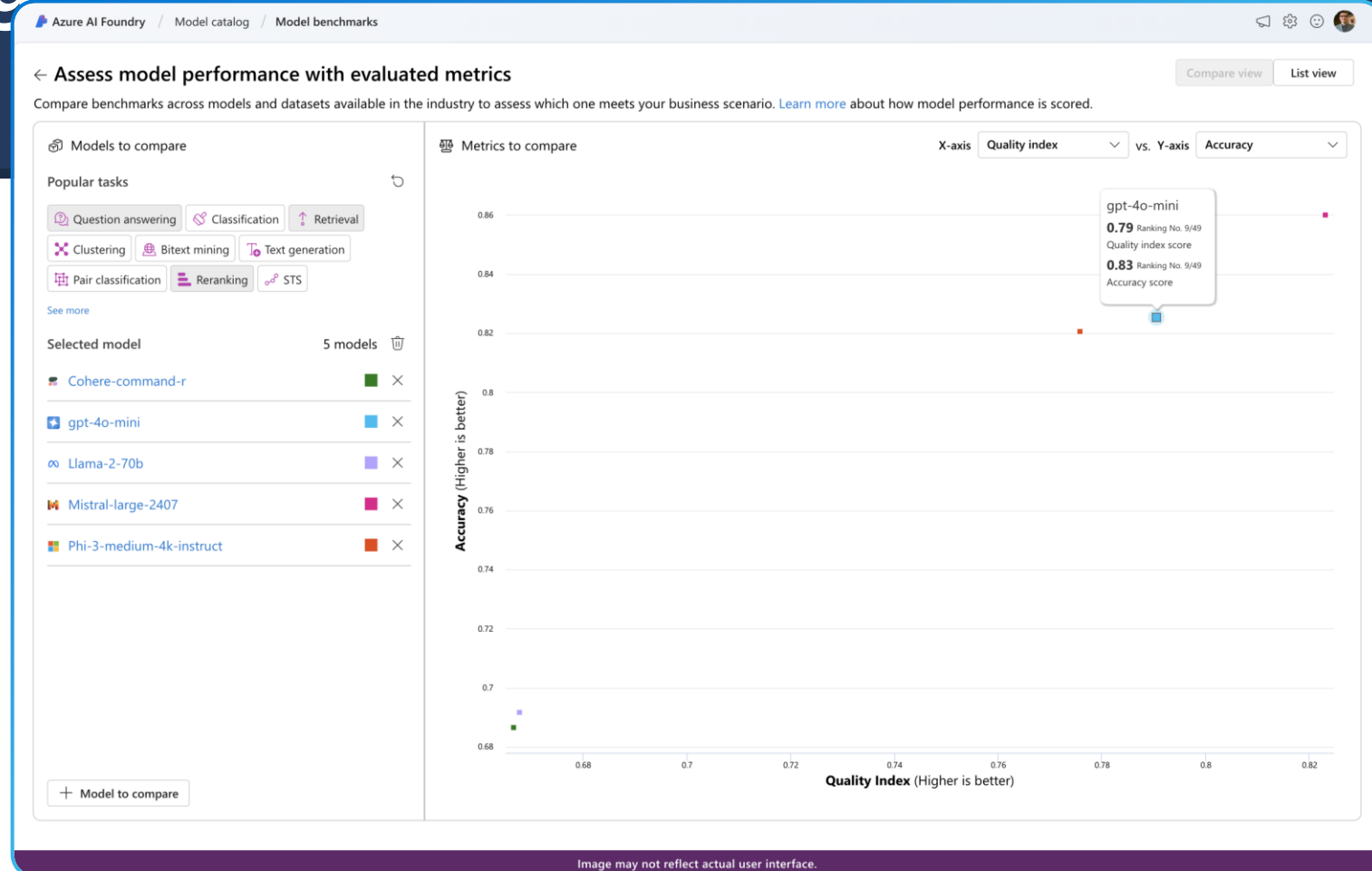


Model benchmarking

Discover models for your
use case

Model Benchmarking enables you to review and compare the performance of various AI models, simplifying the selection process and allowing users to make confident choices with their modeling needs.

- Gain quality, performance, and cost metrics for Azure OpenAI Service, Llama 2 family, Code Llama, and Mistral models
- Access pre-built metrics and benchmark comparison models within the same build, train, deploy environment
- Compute benchmark scores at both the task (dataset) and model levels by utilizing public datasets, yielding a model score for each dataset
- Compare scores of multiple models across datasets and tasks
- Benchmark results originate from public datasets; Azure AI evaluation pipelines download data from original sources, extract prompts from each row, generate model responses, and then compute relevant accuracy metrics



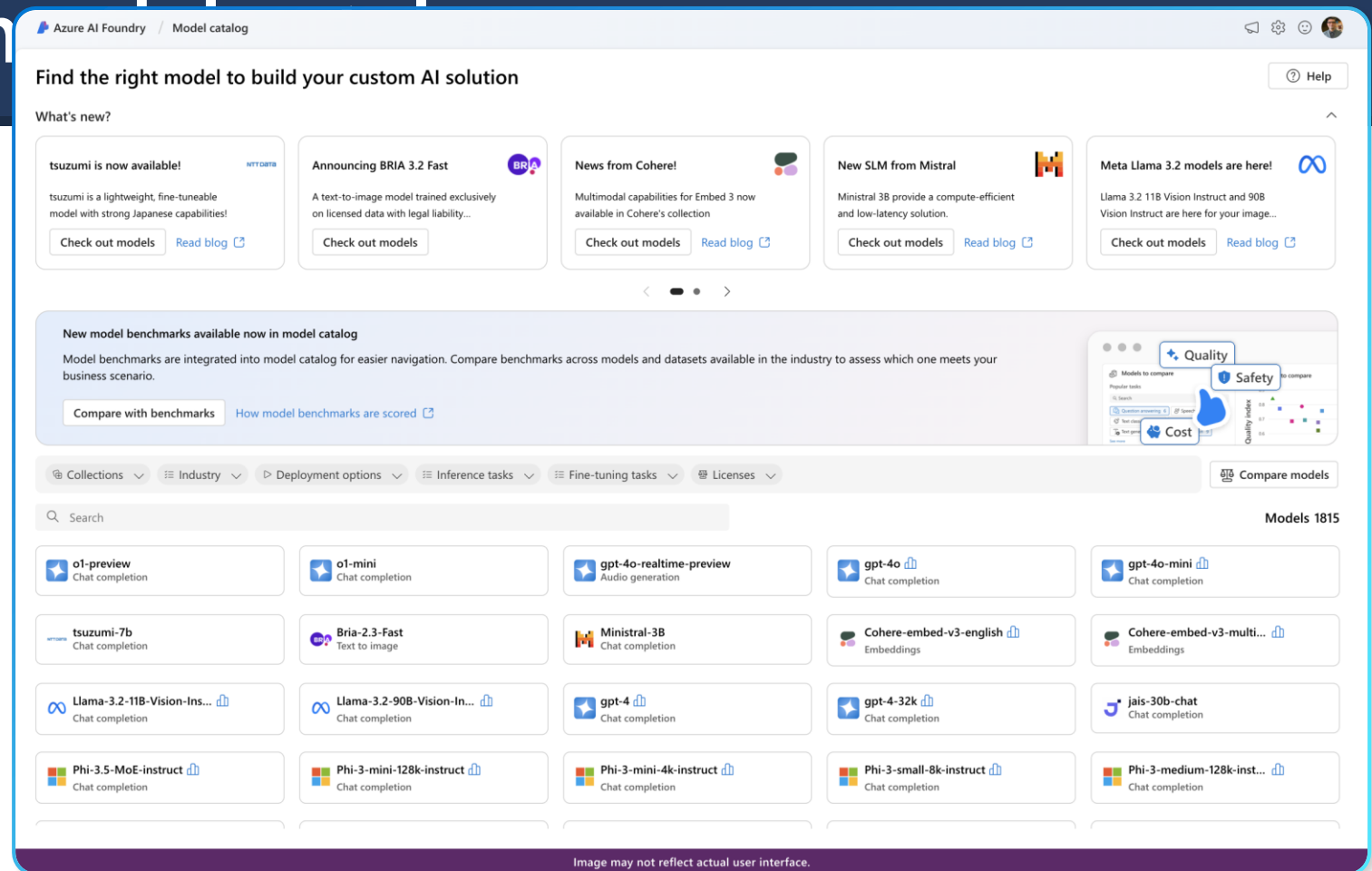
Enable the comparison of models based on accuracy and empower users to make data-driven decisions, ensuring their AI solutions are optimized for the best performance.

Models as a Platform (MaaP) Offering within the new

Discover models for your
use case

Models as a Platform (MaaP) lets you innovate with models from vetted providers using a generative AI model store to explore large foundation models curated by Microsoft, Hugging Face, Meta, and other open-source contributors.

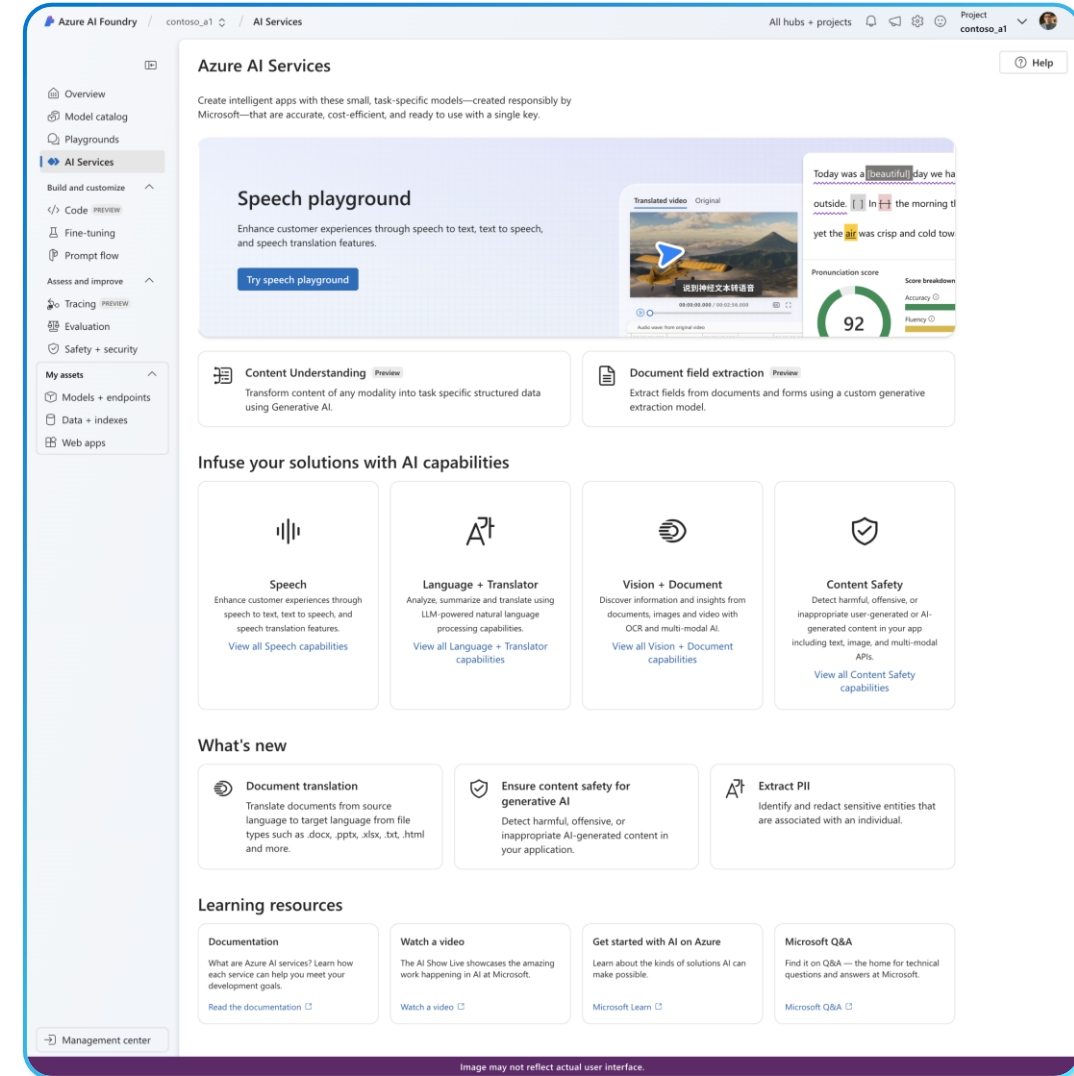
- Explore thousands of large foundation models (LLMs and SLMs) packaged for out-of-the-box usage and optimized for Azure AI Foundry
- Manage your own compute resources with a self-managed GPU infrastructure
- Enhance model performance with highly customized fine-tuning
- Strengthen data security and privacy by controlling the network environment with managed virtual network scenarios
- Access shared compute resources and temporary endpoints for testing, available for seven days and intended for use in testing scenarios
- Industry AI models for Financial Services, Manufacturing, Consumer Good, Mobility



MaaP offers scalable, self-managed hosting for greater control and customization, giving developers flexibly to choose from the most comprehensive selection of open-source generative AI models.

Azure AI Services

Build cutting-edge, market-ready AI applications with out-of-the-box and customizable APIs and multi-modal models



Design intelligent apps with Azure AI services

Leverage out-of-the-box and customizable APIs and multimodal models

Azure OpenAI Service

- Access to powerful AI models
- Scalable development
- Compliance & security
- Integration with other Azure Services

Azure AI Search

- AI enrichment & semantic ranking
- Generative AI content creation
- Vector search for data organization

Azure AI Speech

- Speech to text (including the Whisper model on Azure OpenAI Service)
- Text to speech
- Speech translation
- Speaker recognition

Azure AI Vision

- Image and face analysis
- Custom model training
- Face detection and recognition
- Document text extraction

Azure AI Content Safety

- AI-driven content moderation for enhanced safety
- Customize safety thresholds for diverse user types
- Detect and prevent Jailbreak Risk from XPIA attacks

Azure AI Document Intelligence

- Automated documentation generation
- Documentation quality analysis
- Interactive documentation experiences
- Natural language understanding for documentation

Azure AI Language

- Task-optimized AI models for text analytics
- Custom industry-specific AI for healthcare
- Custom, industry-specific models

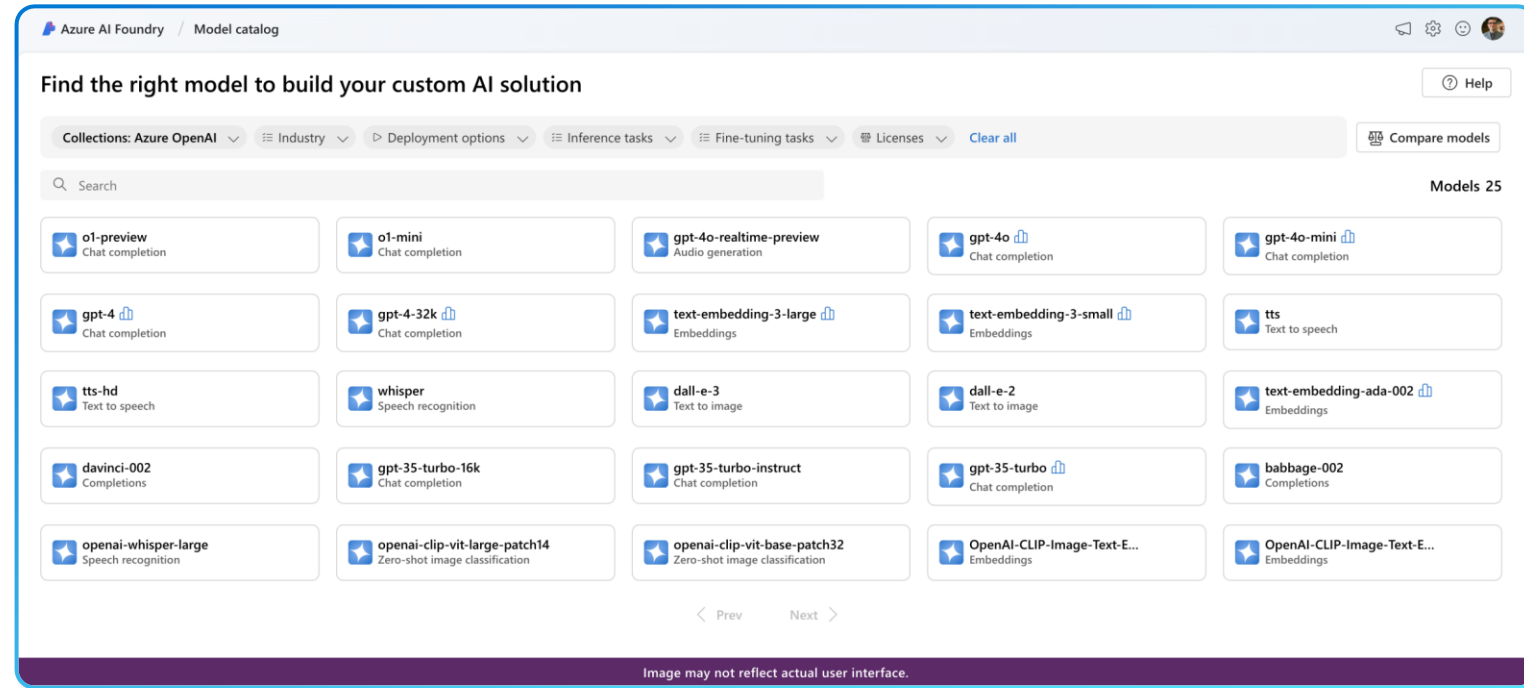
Azure AI Translator

- Multilingual text and speech translation
- Synchronous and asynchronous translation request support
- Native translation of documents and manuals

Azure OpenAI Service (AOAI)

Azure OpenAI Service empowers developers and businesses to harness the power of advanced AI technologies without the need for extensive AI expertise or infrastructure investment.

- Latest innovations with same-day access - Offering the latest models in the Azure OpenAI Service on the same-day with the same features.
- Widest Range of Deployment offerings - Standard, Provisioned and Batch available across 28 different regions, data zones and Global
- Your data is private and secure. Data residency at rest and at processing, and data protection from exfiltration.
- Safety is built-in with tools to responsibly build and implement AI.
- Enterprise Azure promises with private networking, managed identity, customer managed keys, azure monitoring, cost management, and more.
- Developer-Friendly Integration. Seamless integration with GitHub, Visual Studio and Copilot Studio.
- Choice of Models - Over 1800 models available to integrate into your application.



Leverage OpenAI's state-of-the-art models and Azure's cloud platform to build innovative AI-driven applications and solutions across various industries and domains.



Cutting-edge AI models
Pioneering research and advancements
Unlocking new creative possibilities



Secure data and built-in responsible AI
Global availability and flexibility
Developer-friendly integrations

Azure OpenAI Service

Connect your data | Customize your agents | Manage quota & deploy

Reasoning and
multimodal
text, image, audio

o1
o1-mini
GPT-4o
GPT-4o mini

Fine-tuning

GPT-4o
GPT-4o mini

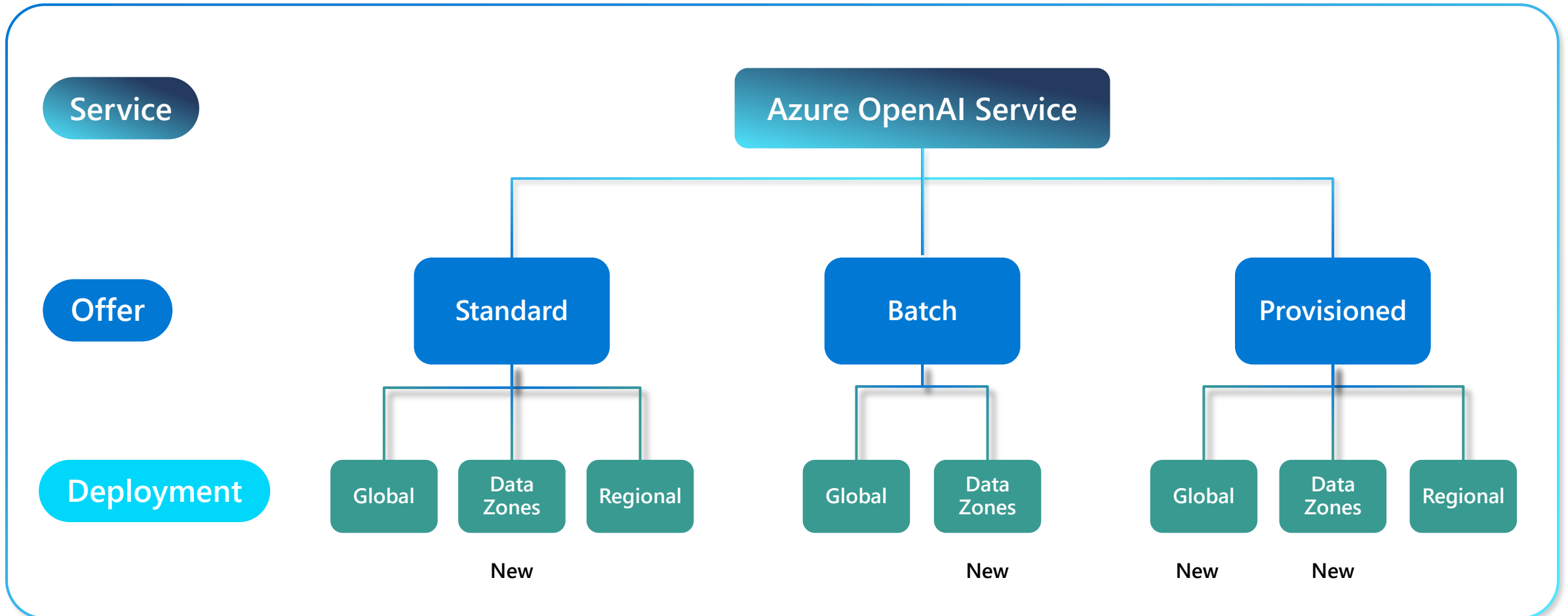
Images

DALL·E 3

Transcription &
translation

Whisper

Azure OpenAI Service Offerings



Azure OpenAI Service model guidance

o1

Multimodal model focused on advanced reasoning and solving complex problems, including math and science tasks.

o1-mini

Smaller, faster, and more cost-efficient version of o1. Performs well at instruction following and small context operations.

GPT-4o

A versatile, multimodal model for text and image processing, non-English languages, and vision tasks.

GPT-4o-mini

A cost-effective version of GPT-4o for lightweight applications.

Azure OpenAI Service model guidance

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Discover your adventure



Join the waitlist to pre-order now

Azure AI Speech

Speech-to-text

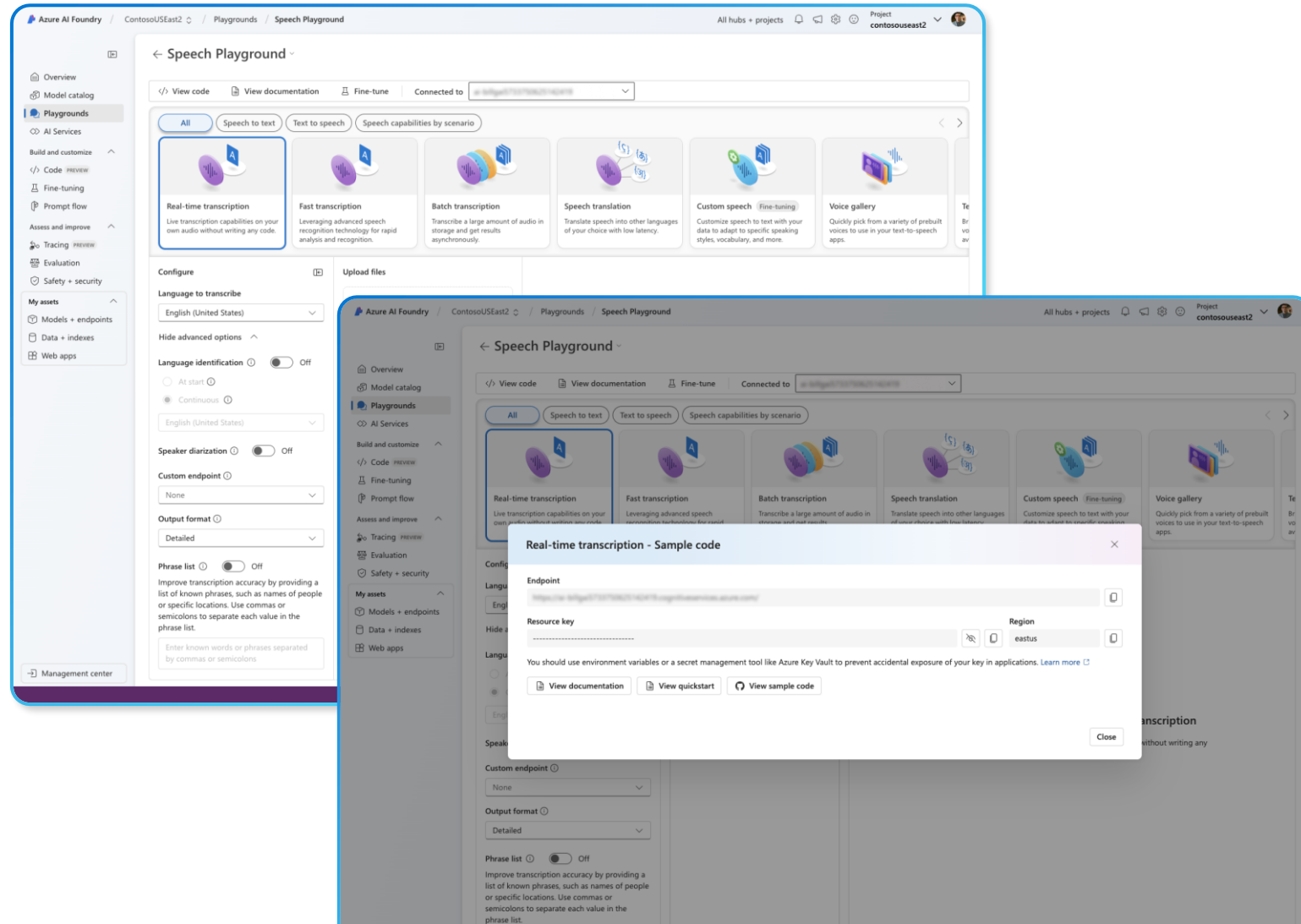
- Contact center – post call analytics and agent assist
- Meeting transcription & insights
- Audio/video transcription & insights
- Dictation
- Caption & subtitles for online and in-person meetings
- Audio and video captions & subtitles

Text-to-speech

- Context-aware, highly expressive HD voices
- Professional voice/brand voice
- Avatars

Speech translation

- Interpreter agent in Microsoft Teams
- Video translation API
- Real-time captions and subtitles
- Audio and video caption & subtitle
- Batch video



Generate insights from documents, text, audio, images, and video

Azure AI Content Understanding

Differentiators:

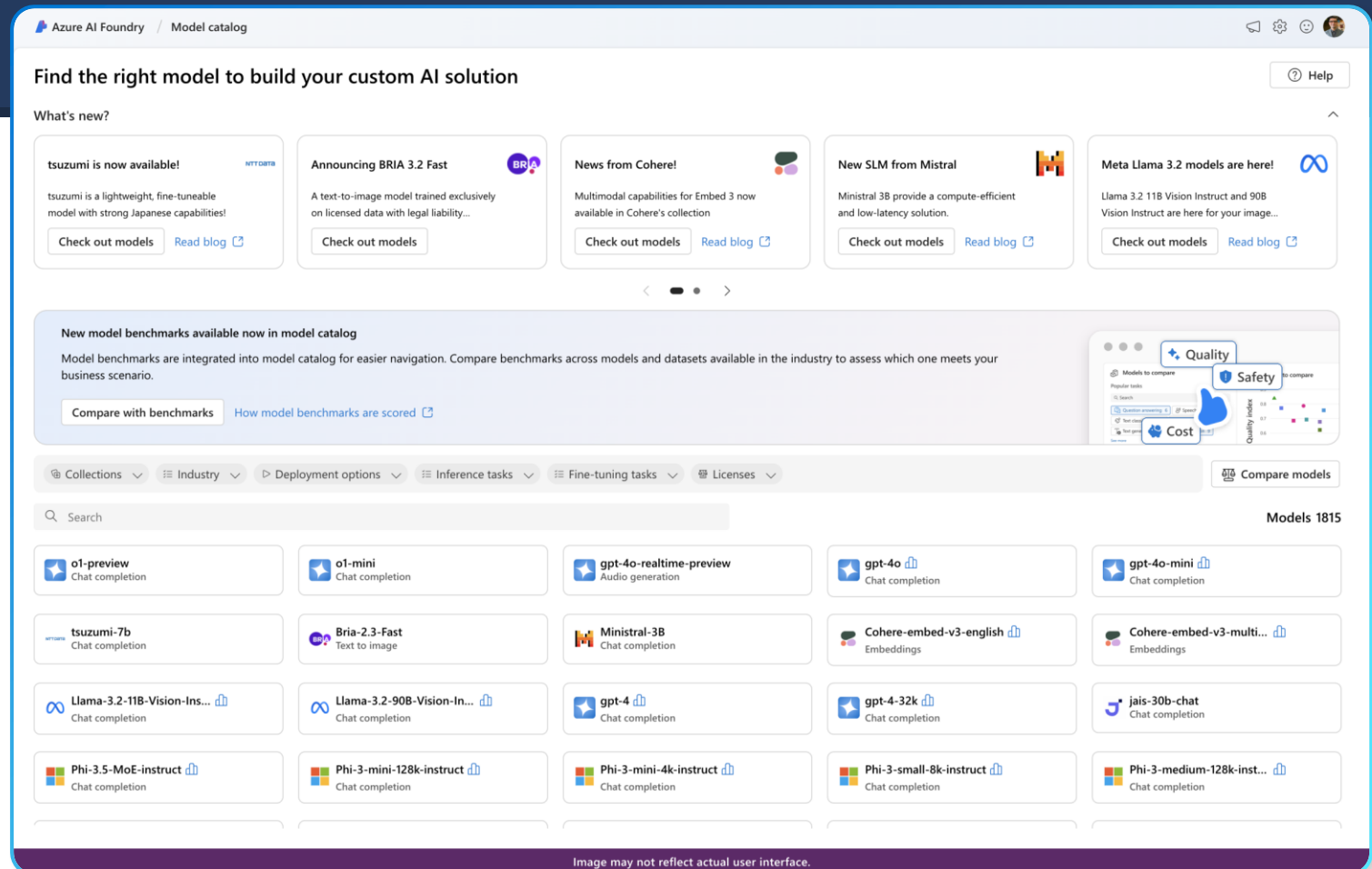
- Streamline multi-modal app development
- Task specific structured output for agentic workflows
- Pre-built templates to speed development



Models as a Service (MaaS) Offering within the model catalog

Models as a Service (MaaS) lets you fine-tune models using serverless APIs, making it easier for generative AI developers to build custom copilots.

- Ready-to-use APIs with pay-as-you-go billing based on tokens
- Customize models with your own data without the need to set up and manage GPU infrastructure
- Easily integrate the latest AI models as an API endpoint to applications
- Integrate with preferred orchestration tools like prompt flow, Semantic Kernel, or LangChain
- Achieve serverless fine-tuning without provisioning GPUs
- Fine-tune Llama 2 with your own data to enhance the model's ability to generate more precise predictions



MaaS provides simplified management with ready-to-use GPU provisioning, lowering costs and reducing barriers to adoption by eliminating complexity.

Models-as-a-Service

Cutting edge models. Ready to use APIs.

Available today

Mistral | Meta Llama | Phi-3 | Core42 | Nixtla | Cohere

New in
preview

Bria AI

Gretel

New in
preview

NTT DATA

Stability AI

AI21

Cohere Rerank

Serverless provisioned offering

Harness your agility to select what's best for your use case with efficiency at scale



Flexible deployment

- Model independent units for any models
- Real-time allocation
- Guided region availability



Scalability

- Default quota in all regions
- Self-serve with auto-approval*



Predictable costs

- Hourly billing + Azure Reservation
- Significant discounts*



Easier access

- Self service
- Try before you buy
- Partner-enabled

*Customers who intend long-term usage of PTUs may receive a significant discount on the hourly rate by purchasing an Azure Reservation for Provisioned.

Flexible deployment options

Serverless API Standard

- Flexible deployment options
- Optimized for low to medium volume workloads
- Pay-per-call flexibility

GOOD FOR:

- ✓ Low to medium volume workloads
- ✓ Intermittent usage scenarios
- ✓ Use cases resilient to throughput variability or slowness when service is busy
- ✓ Global or Regional deployment options with inherent data residency policies
- ✓ Sandbox/test environment

Serverless API Provisioned

- Easy access with high and predictable throughput
- Real-time scoring for large consistent volume
- High volume w/consistent low latency

RECOMMENDED FOR:

- ✓ Production workloads
- ✓ High volume workloads
- ✓ Throughput heavy workloads
- ✓ Latency sensitive scenarios
- ✓ Regional deployments

Bringing Azure AI innovations to developers through their favorite tools and workspaces

Common API for model inference

Azure AI model inference API allows developers to easily switch models and compare performance

Comprehensive AI toolchain and integration across platforms

Complete set of SDKs for models and tools and access from popular IDEs (GitHub and VS Code) and frameworks (LangGraph, Semantic Kernel)

Scalable, secure, and responsible AI operations

Single GPU pool in serving dynamic demands and built-in GenAIOps and safety tooling



Copilot Studio

Low-Code, managed environment for AI business process automation



GitHub



Visual Studio

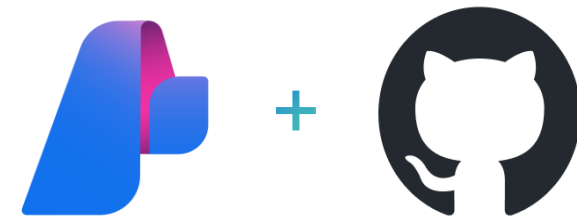
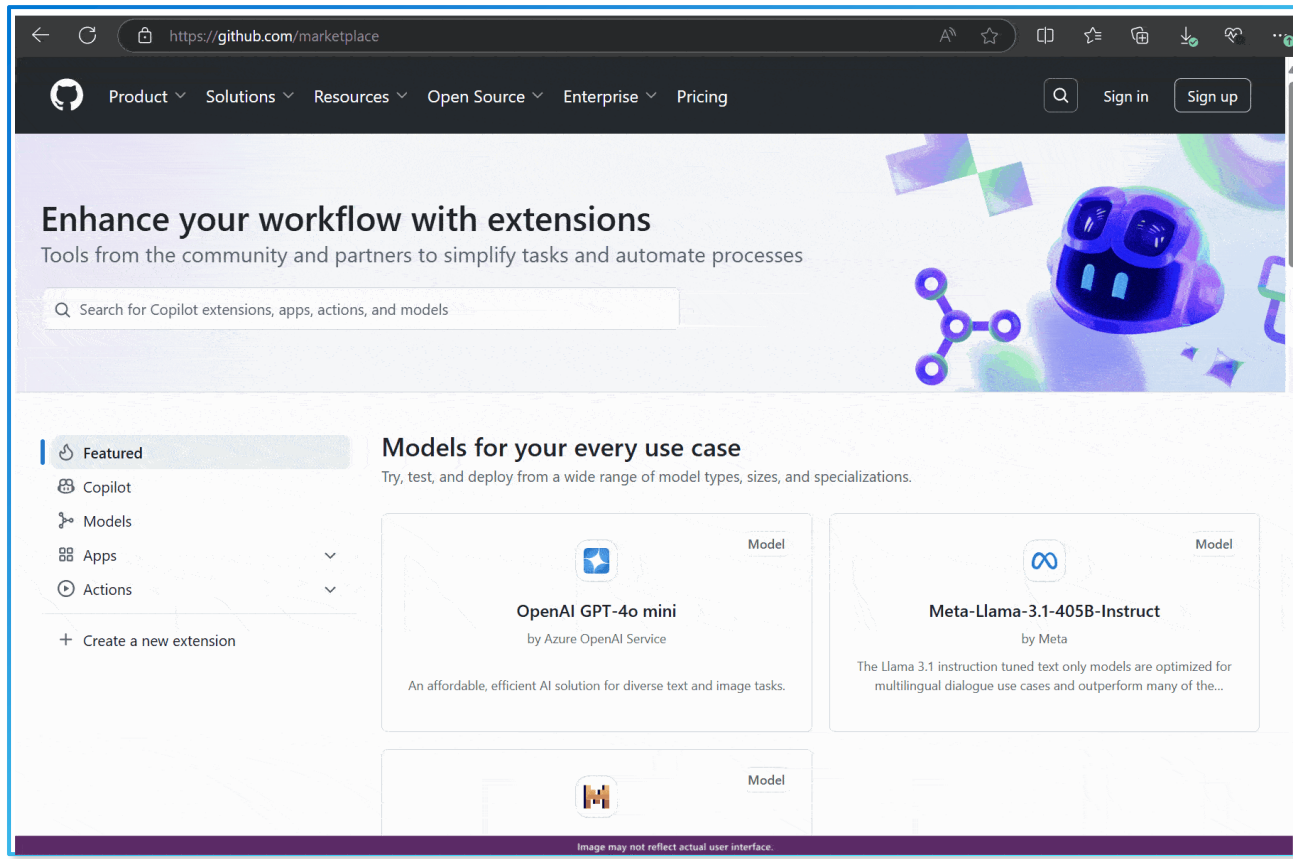
Pro-Code environment for commercial AI solutions



Azure AI Foundry

Comprehensive suite of AI capabilities, models, and tools to design, customize, and manage AI applications and agents

Empowering GitHub Developers to Build with AI



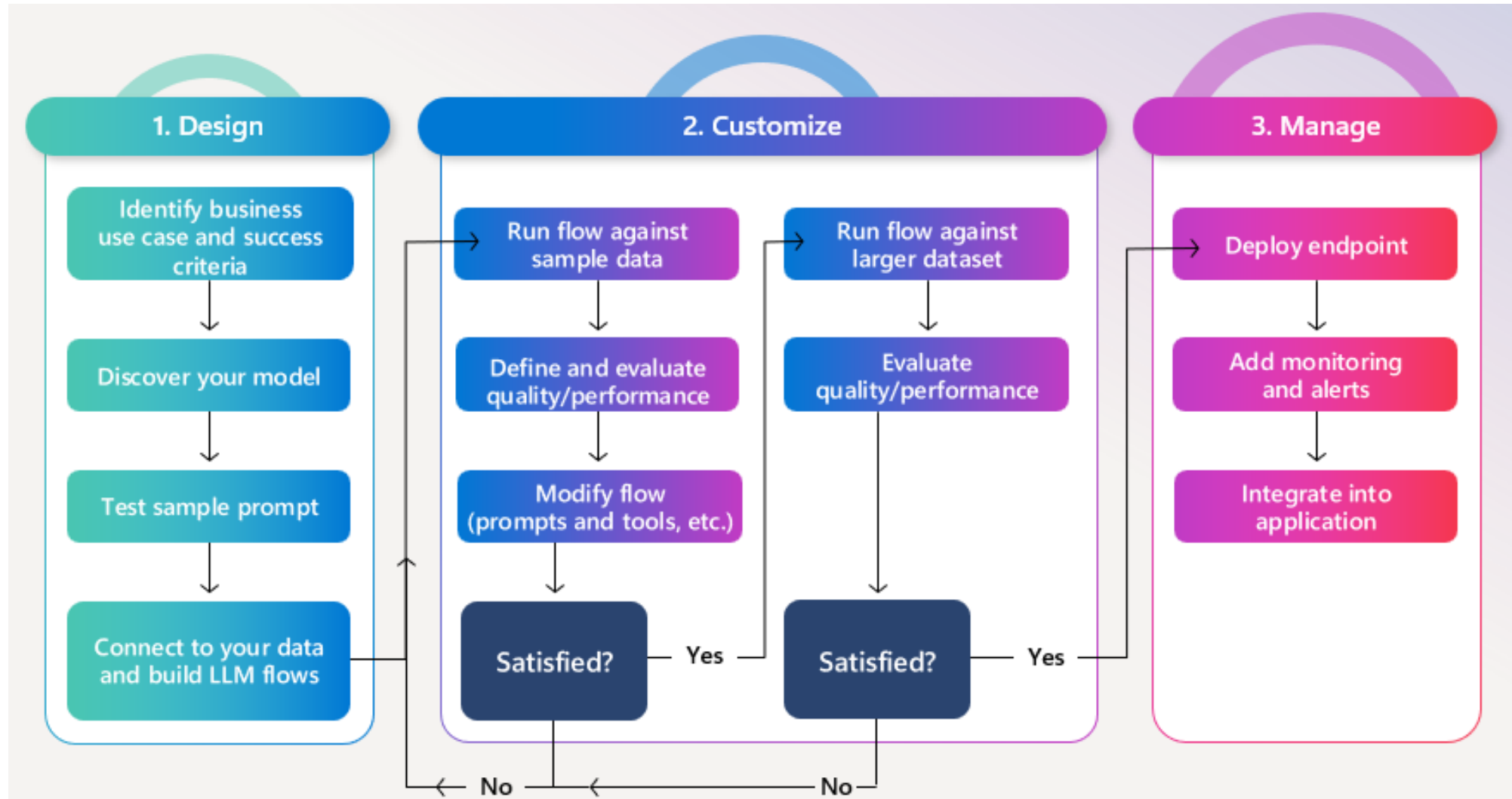
- Get access to a curated list of our top AI models, without leaving GitHub
- Mix, match, and experiment with models in the GitHub playground for free*
- Unlock enterprise-level security and monitoring in production with Azure

Discover in GitHub Models
<https://github.com/marketplace/models>

*Usage limits apply

Complete AI Toolchain

How do we know our results are any good?



Securely developing AI agents requires 4 primary considerations



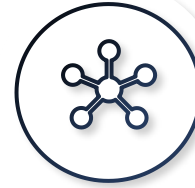
Knowledge

providing agents with the **right context**



Actions

Giving them the tools to complete their tasks



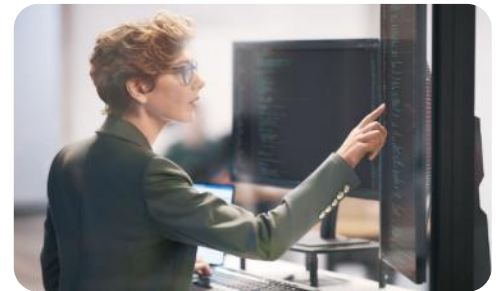
Security

Provide only access to what they should



Evaluation

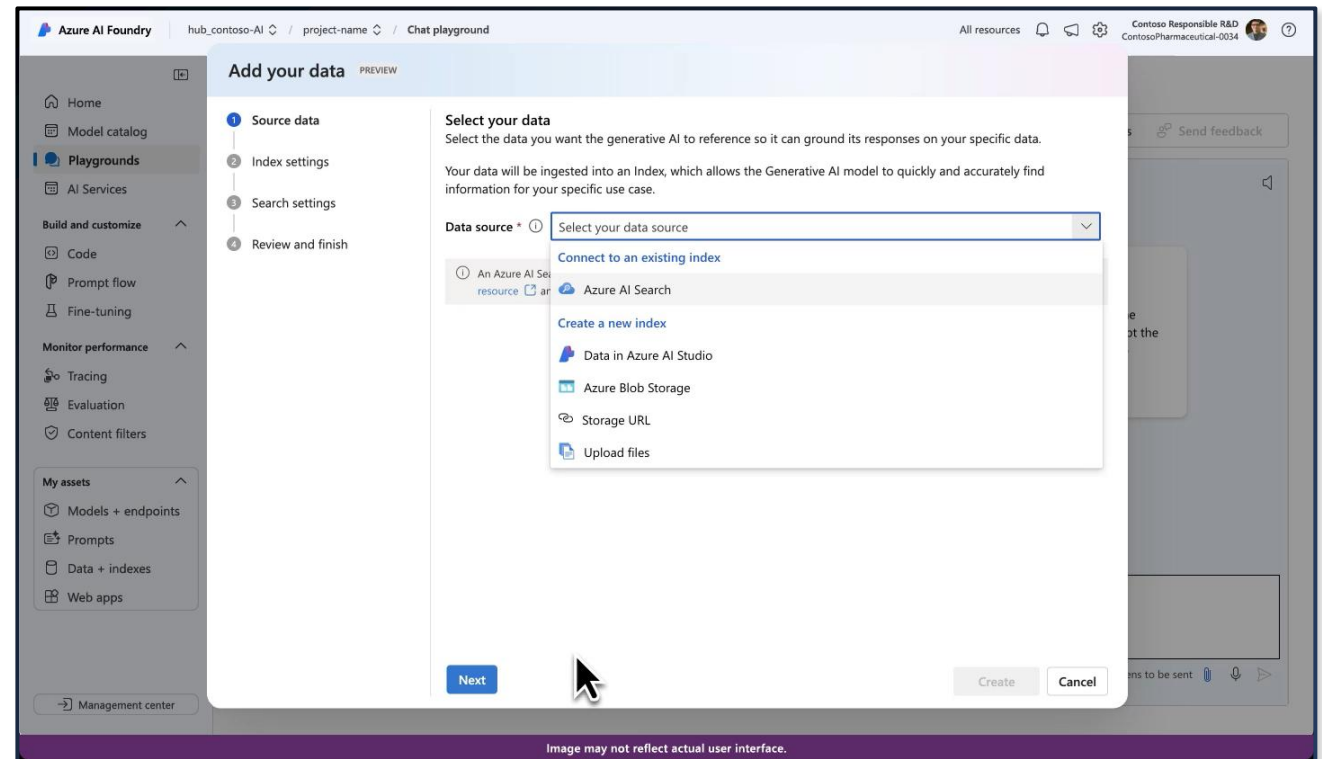
Check they complete their tasks correctly



Customize with an agent toolchain

Differentiate your apps and accelerate development with access to collaborative, comprehensive tooling – wherever you build.

- Customize your models
- Orchestrate and debug AI workflows
- Accelerate AI experimentation



Customize with an agent toolchain

Differentiate your apps and accelerate development with access to collaborative, comprehensive tooling – wherever you build

Customize your models

Train, fine-tune, and distill models using your diverse data sets to innovate and differentiate your AI solutions.

Orchestrate and debug AI workflows

Streamline app development with quick-start AI App Templates, easy-to-use orchestration options, tracing, and debugging tools.

Accelerate AI experimentation

Quickly experiment with models and applications using evaluation frameworks, secure workflows, and real-time monitoring, deploying successful experiments directly to production.

Quick start your AI development



- Quickly deploy applications with the command "azd up"
- Access common starter AI scenarios
- Use GitHub Actions for CI/CD deployment and evaluation to test model performance
- Work consistently and repeatably across the terminal with AZD, your editor/integrated development environment, GitHub Actions pipeline, and more

AI App Templates

The screenshot shows the 'AI app templates' page for 'Azure AI Basic App Sample | Python'. It includes a 'Back to templates' link, a 'Submit feedback' button, and a 'Download template code with the Azure Developer CLI (azd)' button. A code block shows the command: `azd init -t azureai-basic-python`. The page has tabs for 'Overview' and 'Get started'. The 'Overview' tab contains a description of the project, a link to the README on GitHub, and an 'Architecture' diagram. The 'Architecture' diagram shows a flow from 'User input' to 'App Code' (which includes 'Azure AI projects', 'Azure AI model inference', 'Prompt', and 'Azure AI Search (retrieval)') and then to 'Response'. Other components shown are 'Azure Monitor', 'Azure Managed Identity', and 'Azure Application Container (ACA)'. A 'Summary' sidebar on the right lists details like 'Created by', 'Last updated', 'Languages', 'Models', and 'Tags'. At the bottom, there is a table titled 'What's included' listing resources, descriptions, and categories.

Architecture

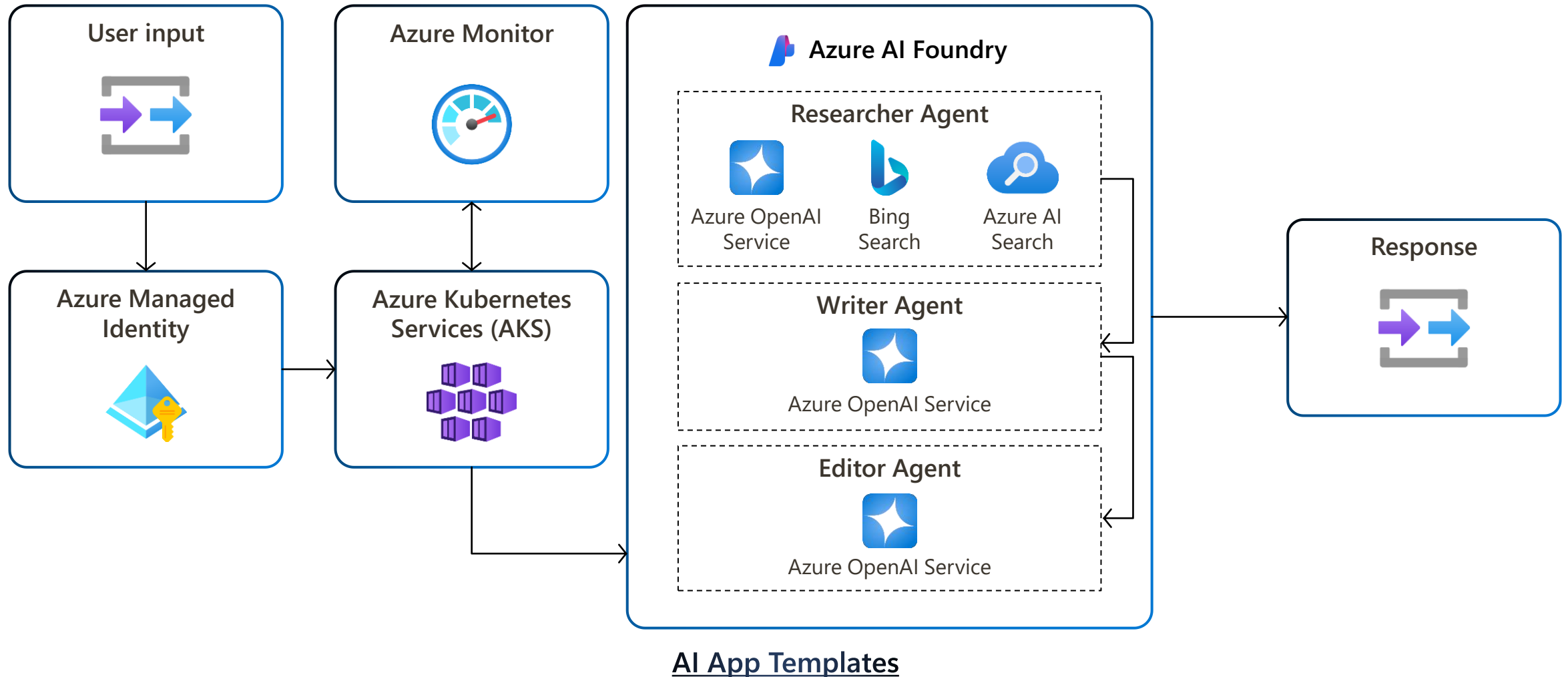
```
graph LR
    UserInput[User input] --> AppCode[App Code]
    AppCode --> Response[Response]
    AppCode --> AzureMonitor[Azure Monitor]
    AppCode --> AzureManagedIdentity[Azure Managed Identity]
    AppCode --> AzureApplicationContainer[Azure Application Container (ACA)]
```

What's included

Resource	Description	Category
Azure Container Apps Azure	Build and deploy modern apps and microservices using serverless containers.	Compute
Azure OpenAI Service Azure	Apply advanced coding and language models to a variety of use cases.	AI + Machine Learning
Azure AI Search Azure	Enterprise scale search for app development.	AI + Machine Learning, Mobile, Web

Image may not reflect actual user interface.

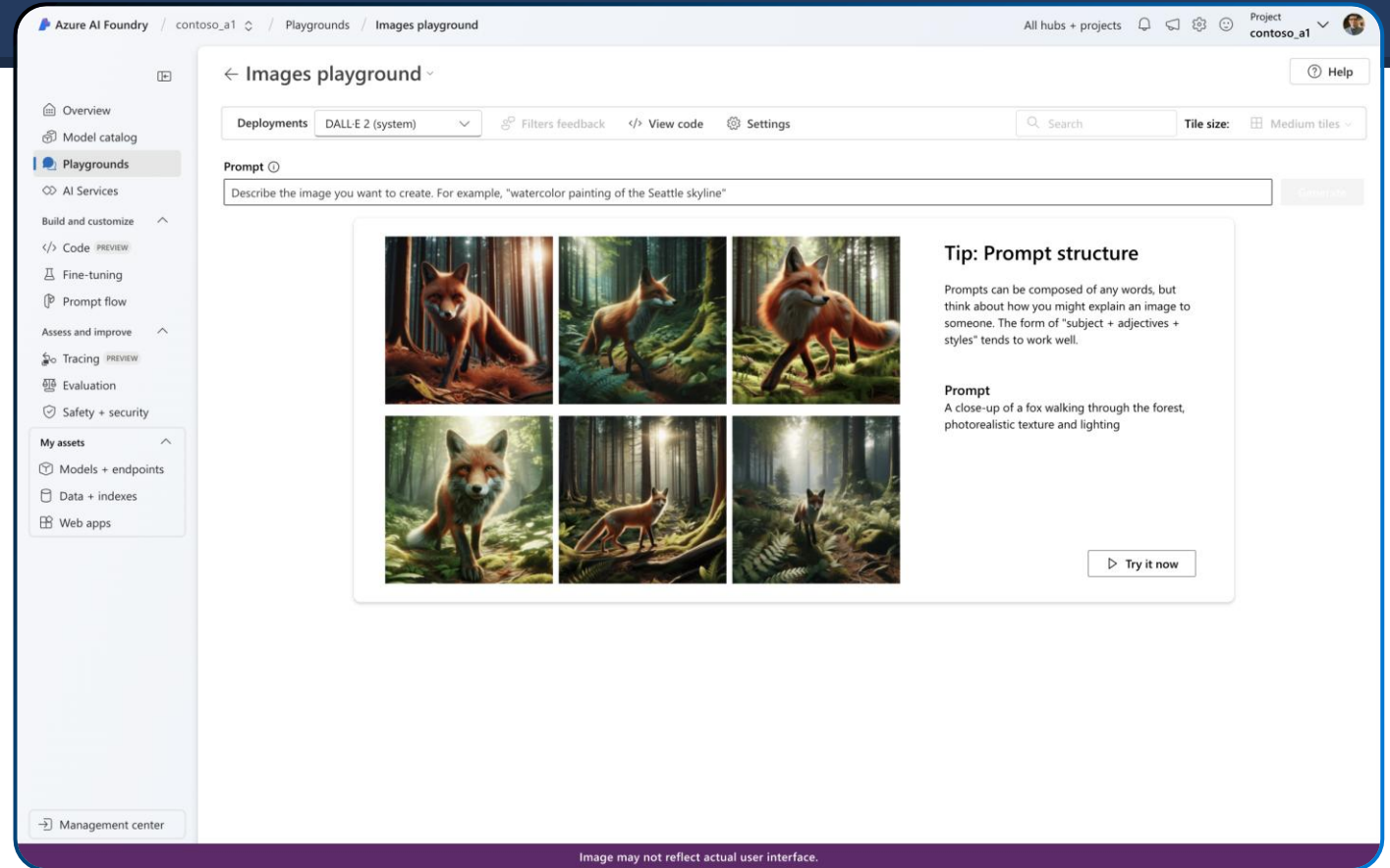
Example AZD template: multi-agent creative writing Copilot



Playground

The playground in Azure AI Foundry offers an environment for developers to experiment with AI models, algorithms, and data.

- Construct system instruction prompts to create your own AI agents and copilots
- Easily connect with indexed data sources to implement retrieval-augmented generation (RAG)
- Leverage prompt samples and templates to eliminate writing long prompts from scratch
- Easily import prompt samples from local or export for local development
- Pair models with Azure AI services APIs to enable richer interactions
- Click "Customize in prompt flow" to further enrich and manage large language model (LLM) workflows

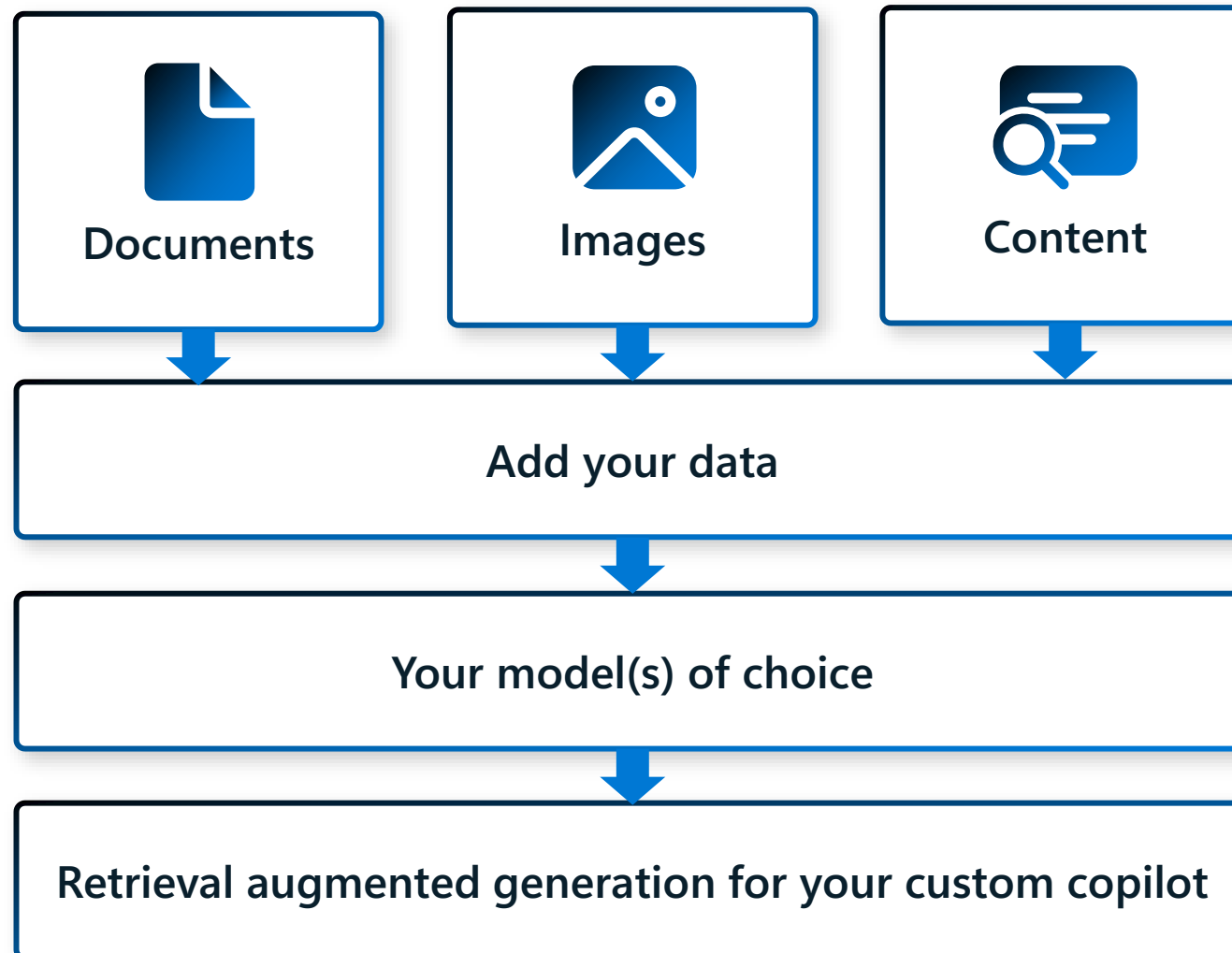


Give developers the freedom to experiment, innovate, and learn without constraints of production environments.

Cutomizing your AI Applications

	RAG	Function/API	Agent
Scenario Example	One well-defined task (Q&A) Known outcomes (answers) Looks for answers in product documentation	Multiple tasks Controlled outcomes (calls) Query an API to validate account number	Complex task Open outcomes Chain multiple apis to solve a problem
Models	Question answering (understanding/summarization)	Intent detection and planning	Multiple
Orchestration	Systematic workflow	Flexible Resilient (Code is the trigger)	Dynamic Self-healing (Agent is the trigger)

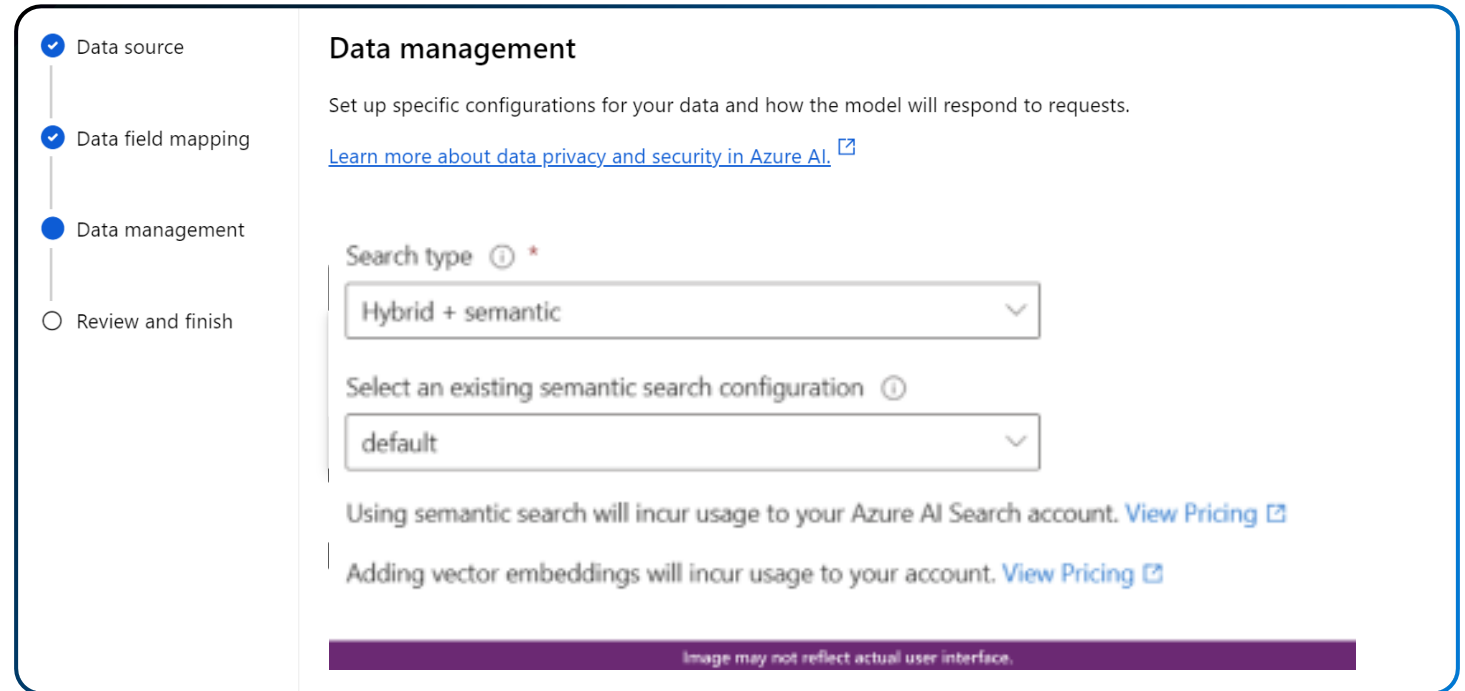
Retrieval Augmented Generation



Azure AI Search

Azure AI Search is a battle-tested retrieval system purpose-built to deliver superior retrieval augmented generation (RAG). Use AI to collect, enrich and surface all aspects of your data.

- Ingest and search across all your data, no matter the type, volume or source
- Streamlined RAG pipeline, from import to chunking to embedding.
- Enterprise-ready foundation cost-optimized for performance and scale.
- Enable hybrid search (vector + keyword) results using RRF (Reciprocal Rank Fusion) and multi-vector retrieval for multi-modal search
- Improve retrieved results immediately with reranking and query rewriting.
- Tailor RAG experiences by utilizing customizable features such as granular access control and relevance tuning



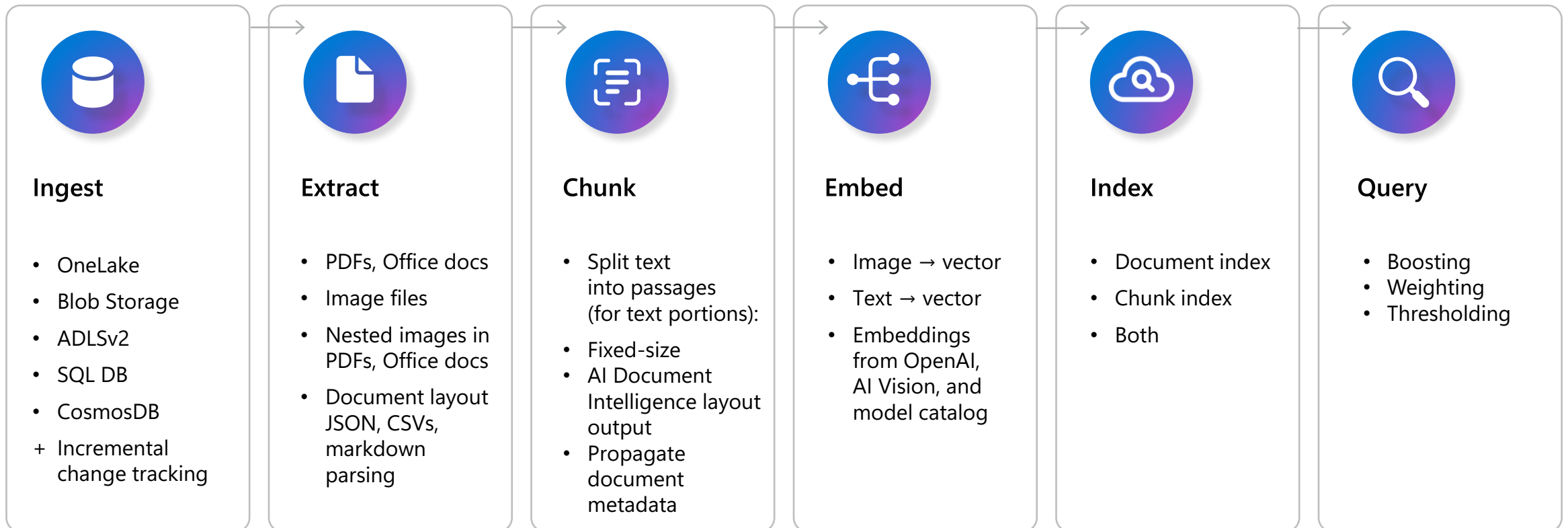
The screenshot shows the 'Data management' configuration page in the Azure AI Search portal. On the left is a vertical navigation pane with four steps: 'Data source' (checked), 'Data field mapping' (checked), 'Data management' (selected), and 'Review and finish' (unchecked). The main content area is titled 'Data management' and includes a subtitle: 'Set up specific configurations for your data and how the model will respond to requests.' Below this is a link: '[Learn more about data privacy and security in Azure AI.](#)'. The 'Search type' dropdown is set to 'Hybrid + semantic'. Below it, the 'Select an existing semantic search configuration' dropdown is set to 'default'. Two informational messages are present: 'Using semantic search will incur usage to your Azure AI Search account. [View Pricing](#)' and 'Adding vector embeddings will incur usage to your account. [View Pricing](#)'. A purple banner at the bottom states: 'Image may not reflect actual user interface.'



Easily enable search best practices through the UI: like being able to select hybrid search + reranking in one click.

Streamlined RAG data pipeline

End to end, integrated data prep and processing



Retrieval Augmented Generation

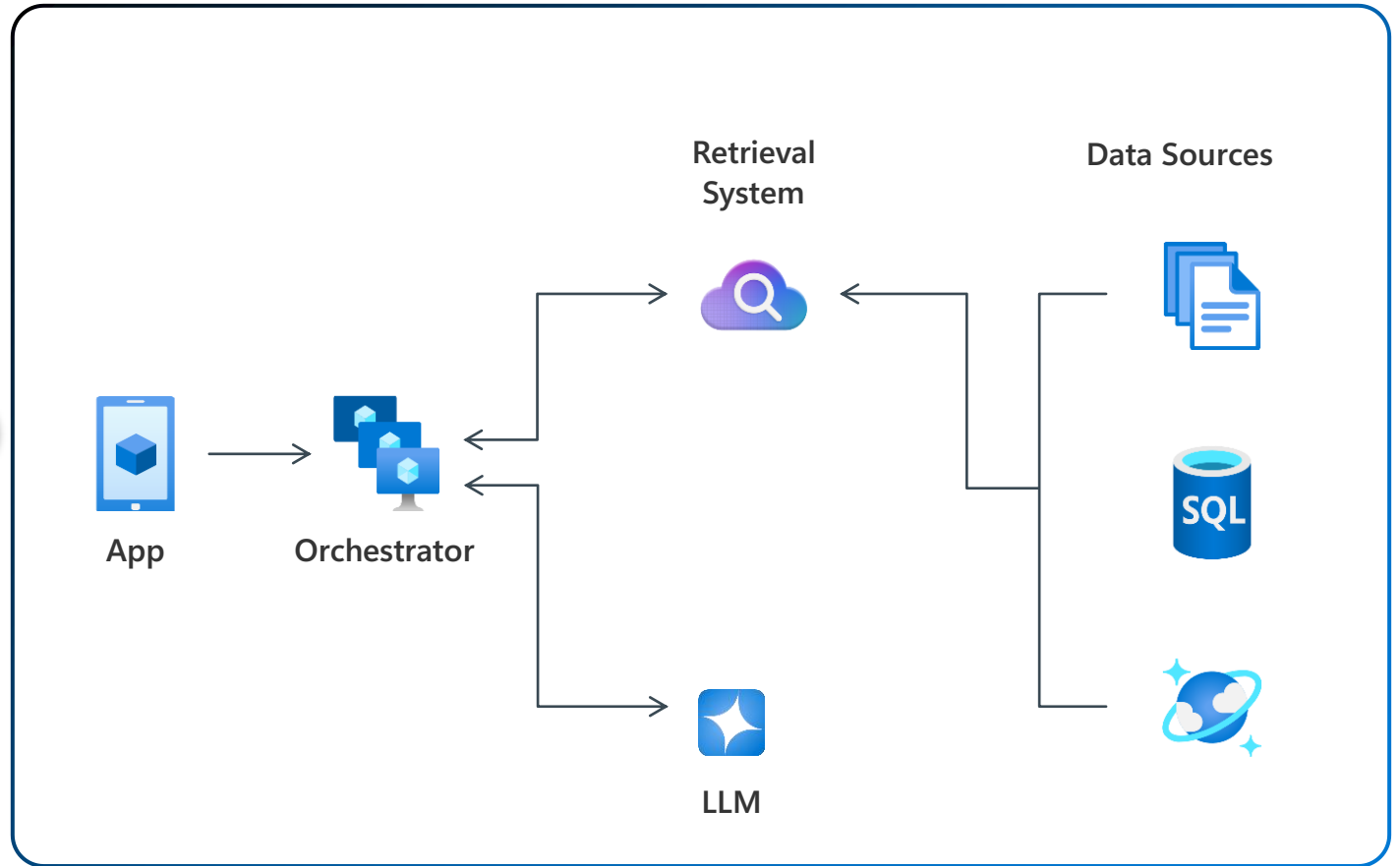
Combine reasoning with knowledge

Elements

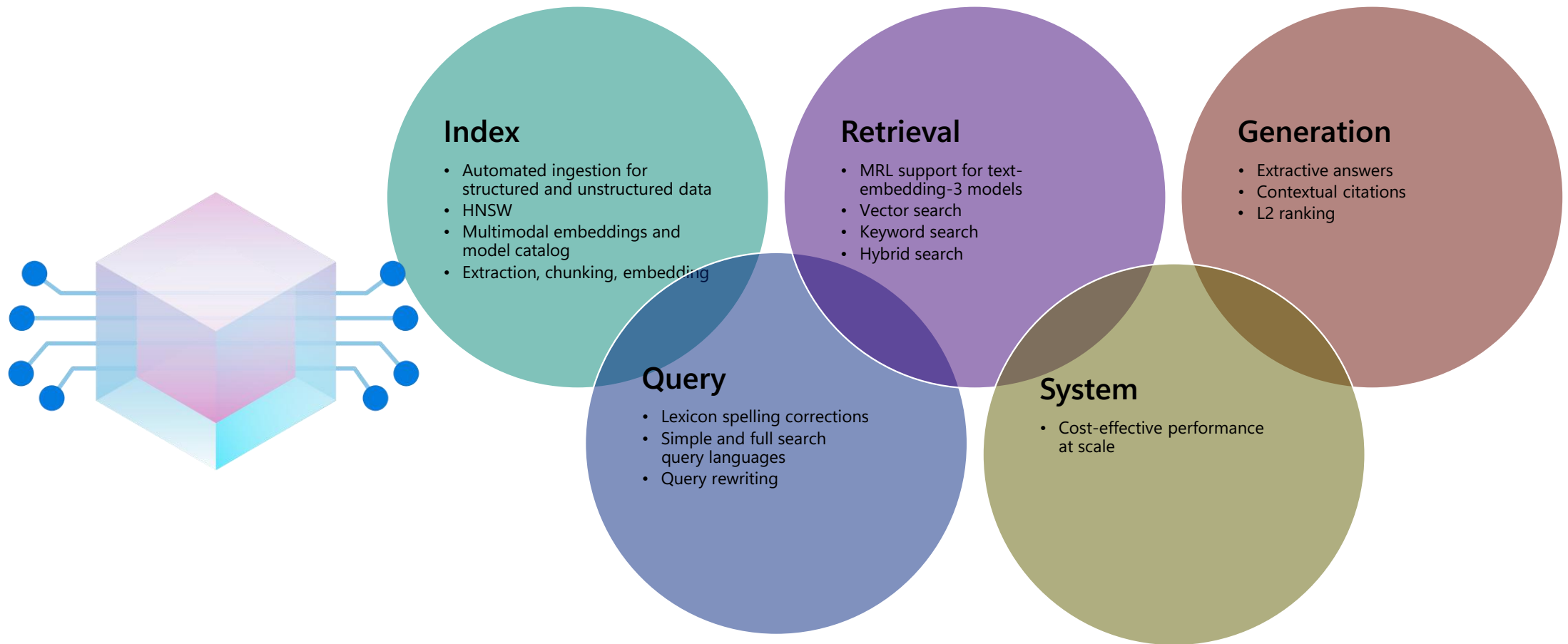
- Retrieval system collects externalized knowledge
- Language model provides reasoning to provided knowledge and prompt

Techniques retrieval system brings

- Querying: spelling corrections, multi-lingual query support
- Retrieval: keyword, vector, hybrid search
- Augmentation: L2 ranking,
- Generation: Extractive answers, citations
- Indexing: chunking, embedding, automated data import



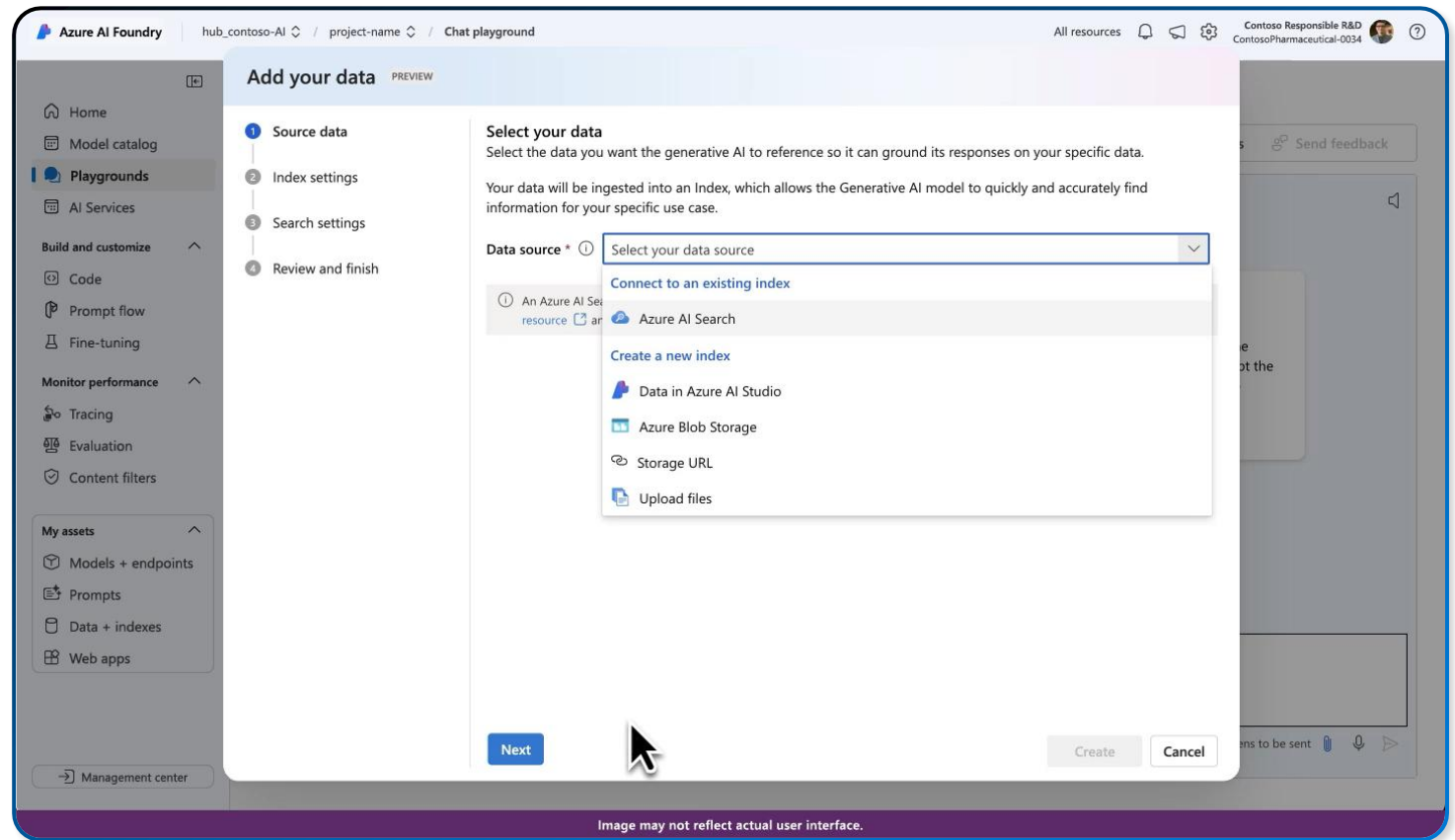
Azure AI Search features across the RAG pipeline



Integrate structured & unstructured data



- **Azure AI Search**
- **Blob storage**
- **Local files/folders**
- **Storage URLs including OneLake in Microsoft Fabric**
 - Azure Databricks
 - S3 buckets via Amazon S3 Shortcuts



Use embeddings from the Azure AI Foundry model catalog

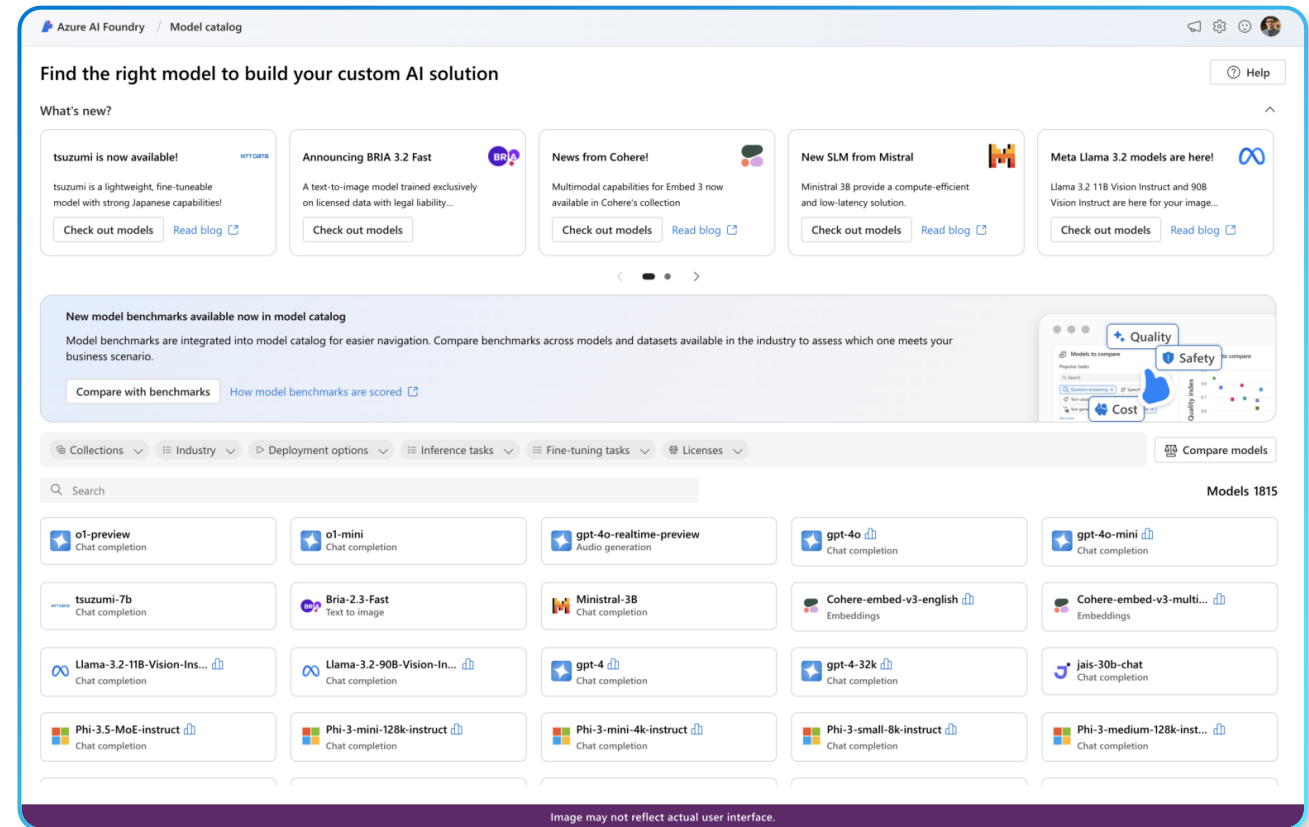
Catalog includes embedding models

Deploy in Azure AI Foundry, connect to Azure AI Search for index/query-time vectorization

Includes models that use narrow types

Example:

Cohere's RAG-focused model "Cohere-embed-v3-English", supports float, int8, binary



RAG for GitHub Models

Free. Frictionless. Just add data.

Seamless playground
experience

Auto-provisioned AI
Search index

Full-featured retrieval
system

Advanced RAG
directly from code

Chat vs. Agent frameworks

Chat	Agents
Lightweight and powerful	Stateful (inbuilt conversation state management)
Inherently stateless	Access persistent threads
	Access files in several formats. API handles chunking, embeddings, storage and creation, and implementing vector search
	Automatic management of the model's context window
	Access multiple tools in parallel (up to 128 tools per assistant) including code interpreter
	Build your own function calling

What does an agent do?



Reason over a provided business process

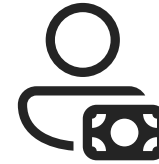


Retrieve context to complete the process

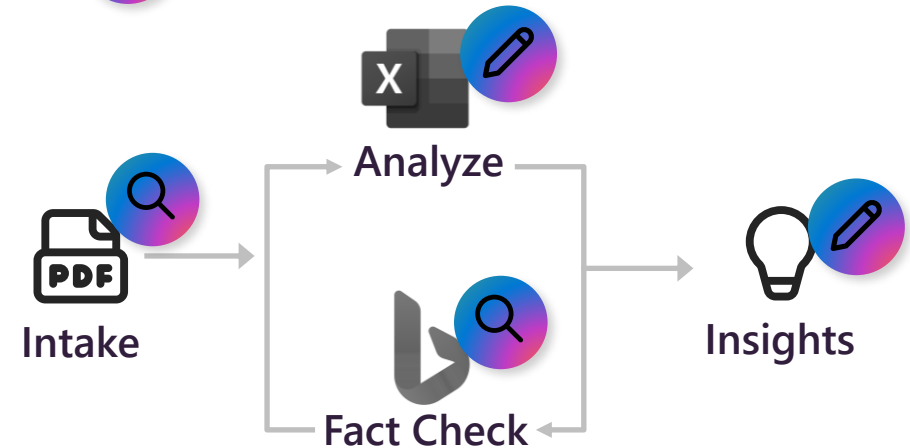


Act on behalf of the user

Financial Analyst

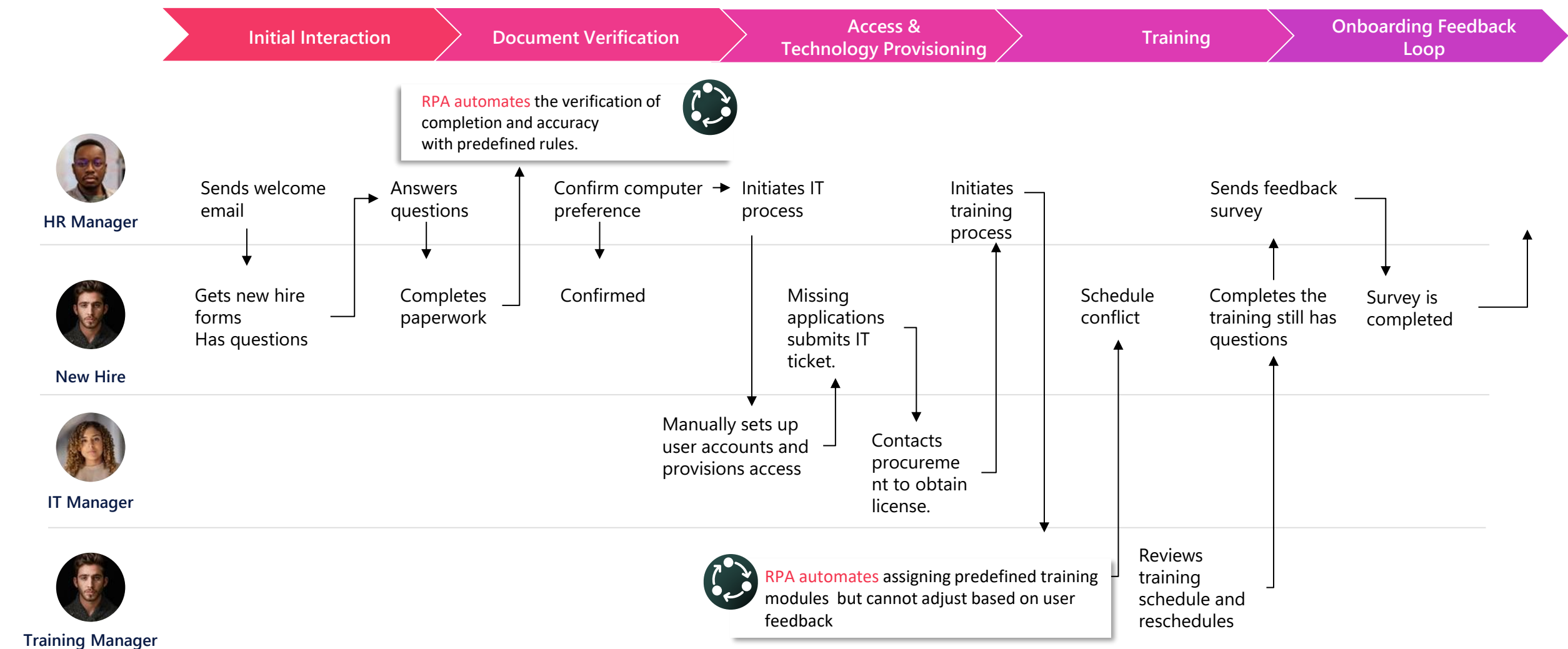


Analysis process orchestration



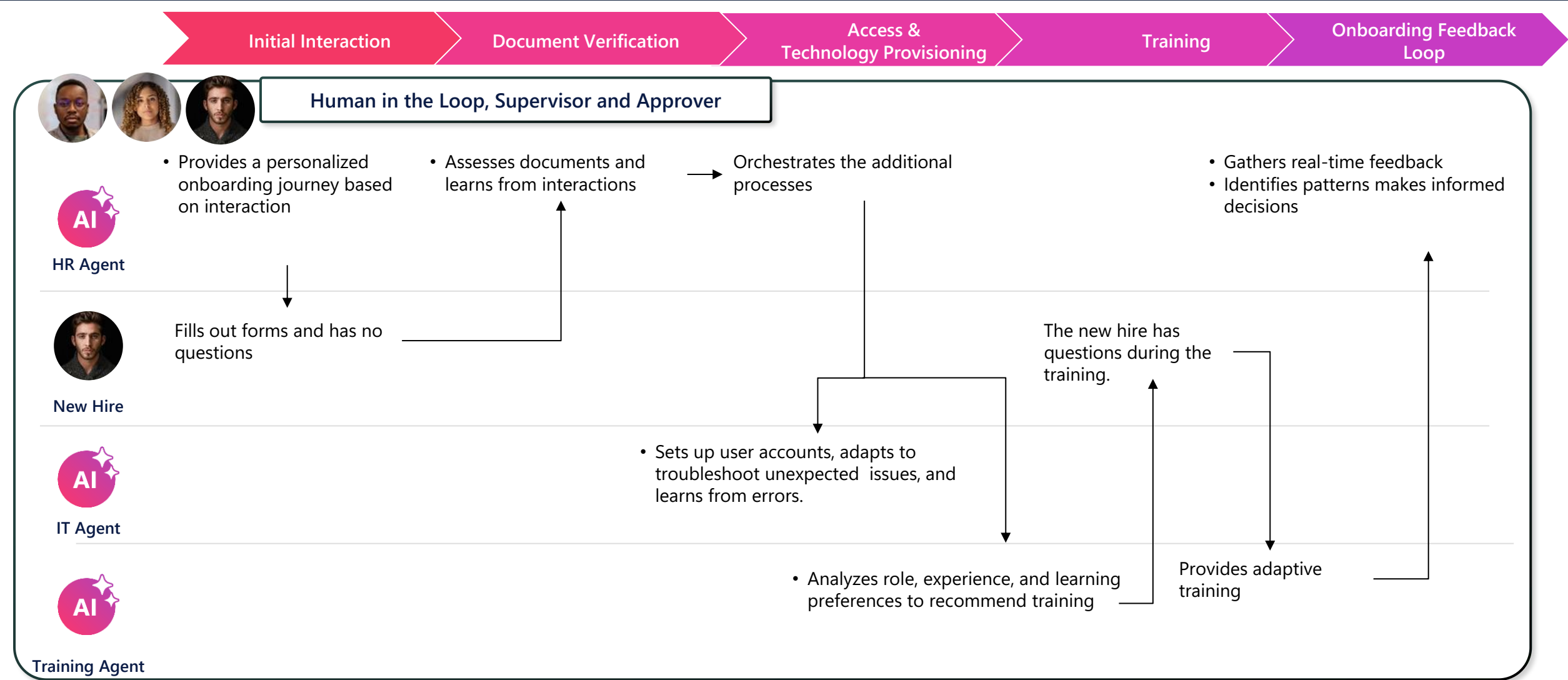
TODAY

Existing solutions can only automate very specific tasks that have clear inputs and outputs



TOMORROW

With AI Agents, these steps can be automated for the first time, while keeping human in the loop.



Azure AI Agent Service

Securely build, deploy, and scale AI agents with ease

Rapid development
and automation

Extensive data
connections

Flexible model
selection

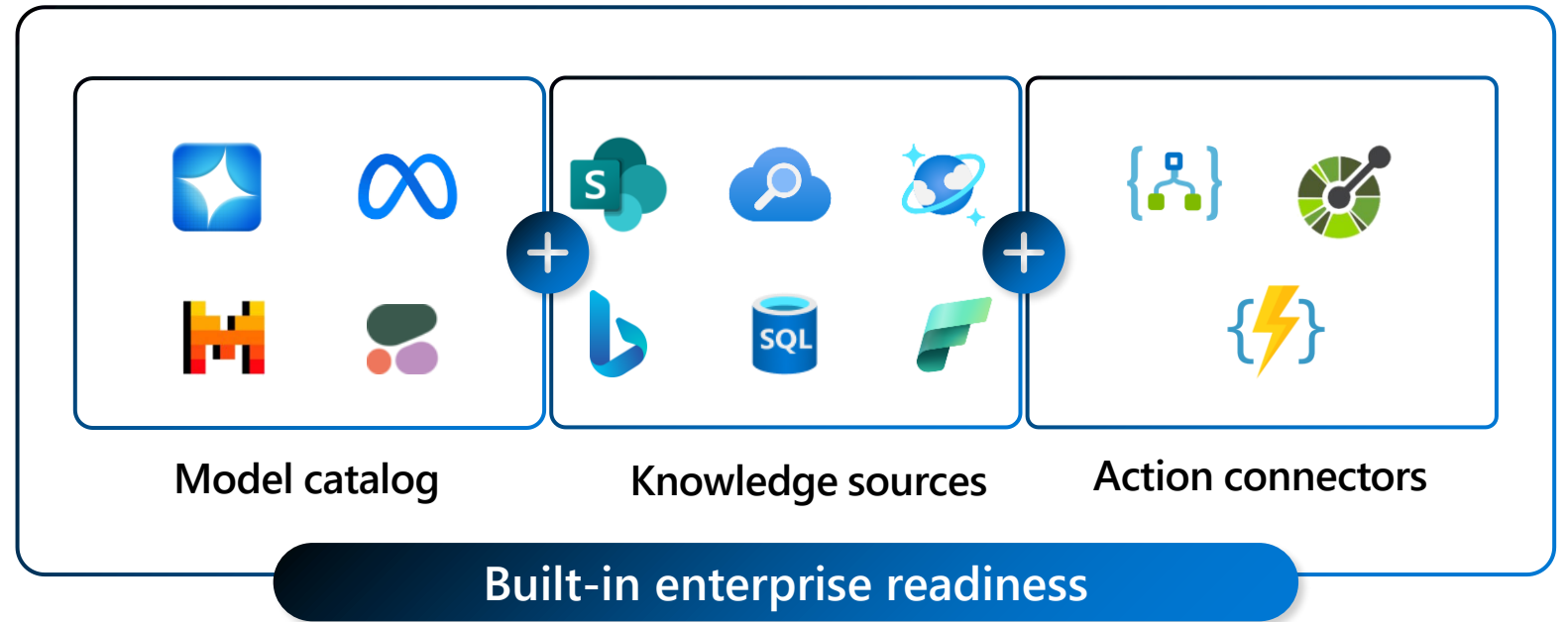
Enterprise-grade
security

[AI.Azure.com](https://ai.azure.com)

Deploy and scale AI agents with ease

Azure AI Agent Service:

- Enterprise-grade security
- Extensive data connections
- Flexible model selection
- Rapid development and automation



Azure AI Agent Service

Built-in enterprise readiness

BYO-file storage
(coming soon)

BYO-search index

OBO Authorization Support

Enhanced Observability



Model Catalog



Azure OpenAI Service
(GPT-4o, GPT-4o mini)

Models-as-a-Service



Llama 3.1-405B-Instruct



Mistral Large



Cohere-Command-R-Plus



Extensive Ecosystem of Tools

Knowledge



Microsoft Fabric*



SharePoint*



Grounding with Bing Search



Azure AI Search



Files (local or Azure Blob)



Your own licensed data



File Search



Code Interpreter

Actions



Azure Logic Apps



OpenAPI 3.0 Specified Tools



Azure Functions

Deploy agents with Azure AI Foundry

Orchestrate them together with
AutoGen and **Semantic Kernel**



State-of-the-art
research SDK



Production-ready
and stable SDK

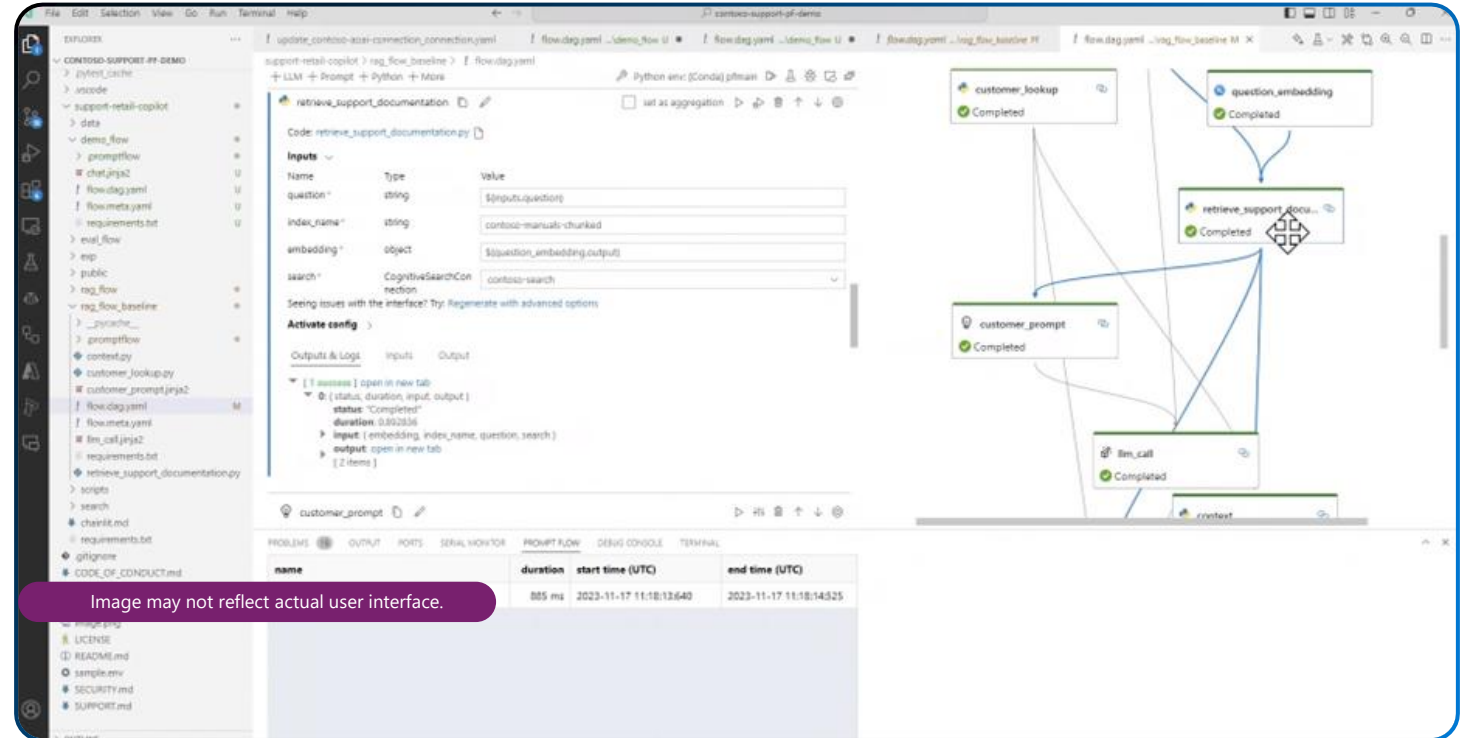
Ideation

Production

Extended tools and frameworks

Prompt flow offers a developer-friendly and easy-to-use code first experiences through SDK, CLI, and VS code extension, allowing seamless integration with open-source frameworks and CI/CD pipelines for automation.

- Use your favorite frameworks and editors that allow you to work in your preferred code environments, whether simplified UI or Visual Studio Code/IDE
- Initially build flows with open-source frameworks like LangChain or Semantic Kernel, and use prompt flow to scale the experiments
- Define flows in YAML format, which can stay aligned with the references source files in a folder structure for flow versioning in code repository
- Easily integrate with existing CI/CD pipelines through GitHub Actions and Azure DevOps for GenAIOps
- Smoothly transition from local to cloud by exporting flow folders to local, or uploading folders to the cloud for further authoring, testing, and deployment



Streamlines the development lifecycle of generative AI app with version-controlling, CI/CD integration, and enhanced collaboration among team members.

Code-first GenAIOps with developer tools



Use code to define flow

- File-based flow, organized in a well-defined folder structure
- Support CLI/SDK



Smooth transition between cloud and local

- Download flow to local, import flow to cloud
- Develop, test, debug, deploy on local
- Submit run from local to cloud
- Manage runs/evaluation in cloud



Integrate with OSS frameworks

- LangChain, Semantic Kernel, AutoGen



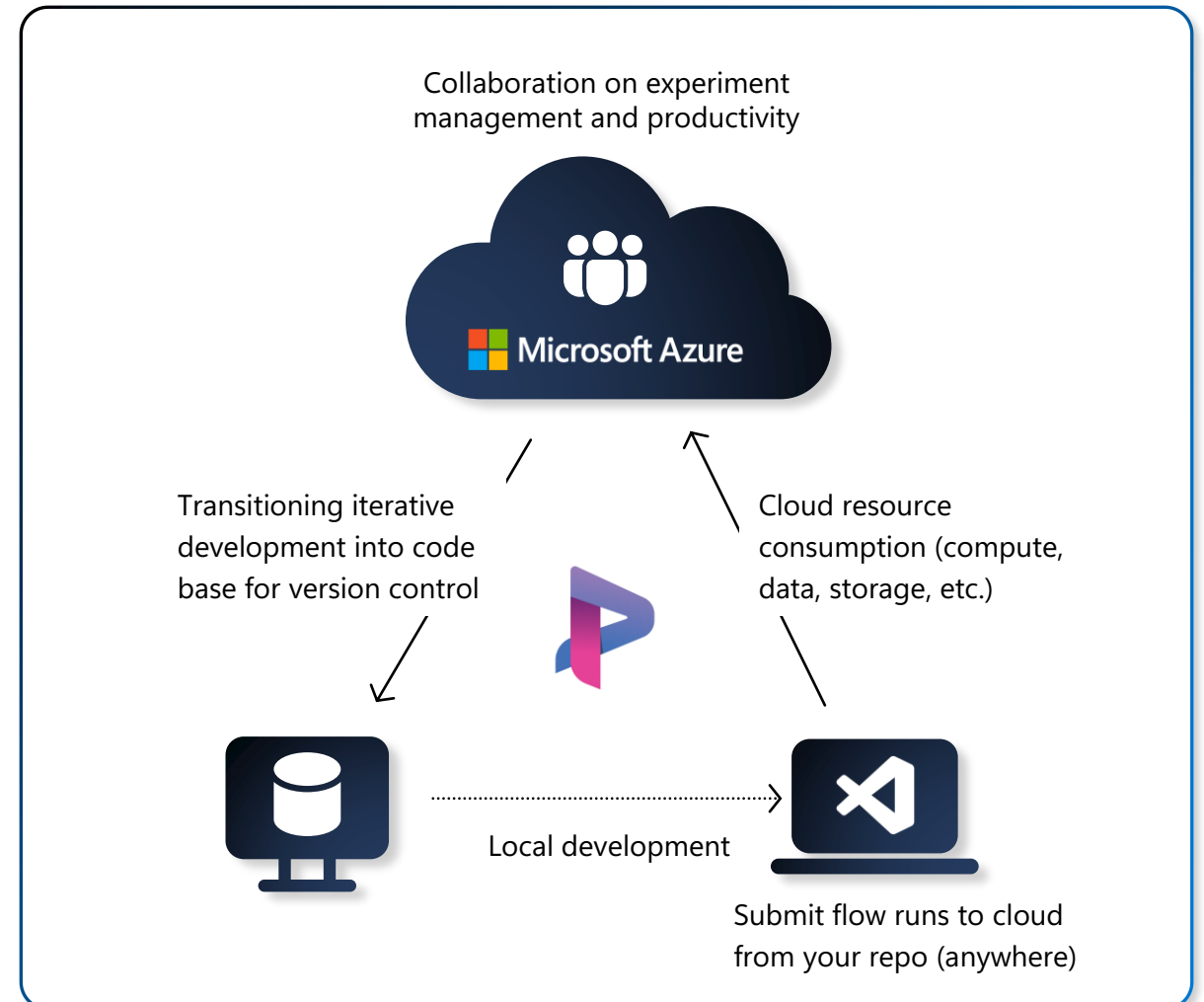
Automate with CI/CD pipelines

- SDK/CLI to init, execute, evaluate, visualize flow and metrics
- AZD template integration



Local development with VS Code Extension

- Flow editor
- Local connection management
- Tracing and run history



Evaluation best practices



Multiply your
datasets
and metrics

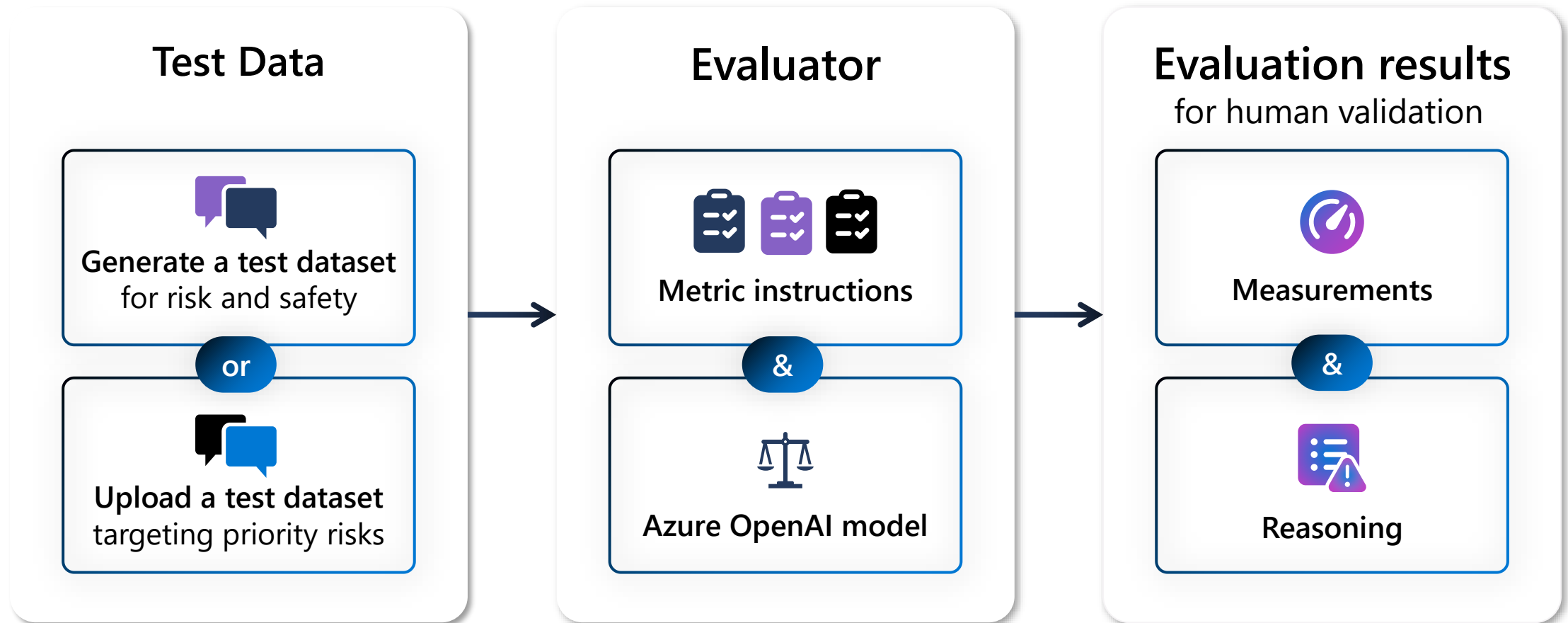


Evaluate
systematically
using GenAIOps



Review results
as a team, with
multiple points
of view

Assess your app with Azure AI Foundry



Automated evaluation in Azure AI Foundry

Text

Quality

AI-assisted metrics

Groundedness

Coherence

Fluency

Relevance

Retrieval score

Similarity

NLP metrics

F1 score, BLEU, ROUGE,
GLEU, METEOR

Text + Image

Risk & safety

AI-assisted metrics

Hate and unfairness

Sexual

Violence

Self-harm

Protected materials

Text

Risk & safety

AI-assisted metrics

Direct attack jailbreak

Indirect attack jailbreak

Text

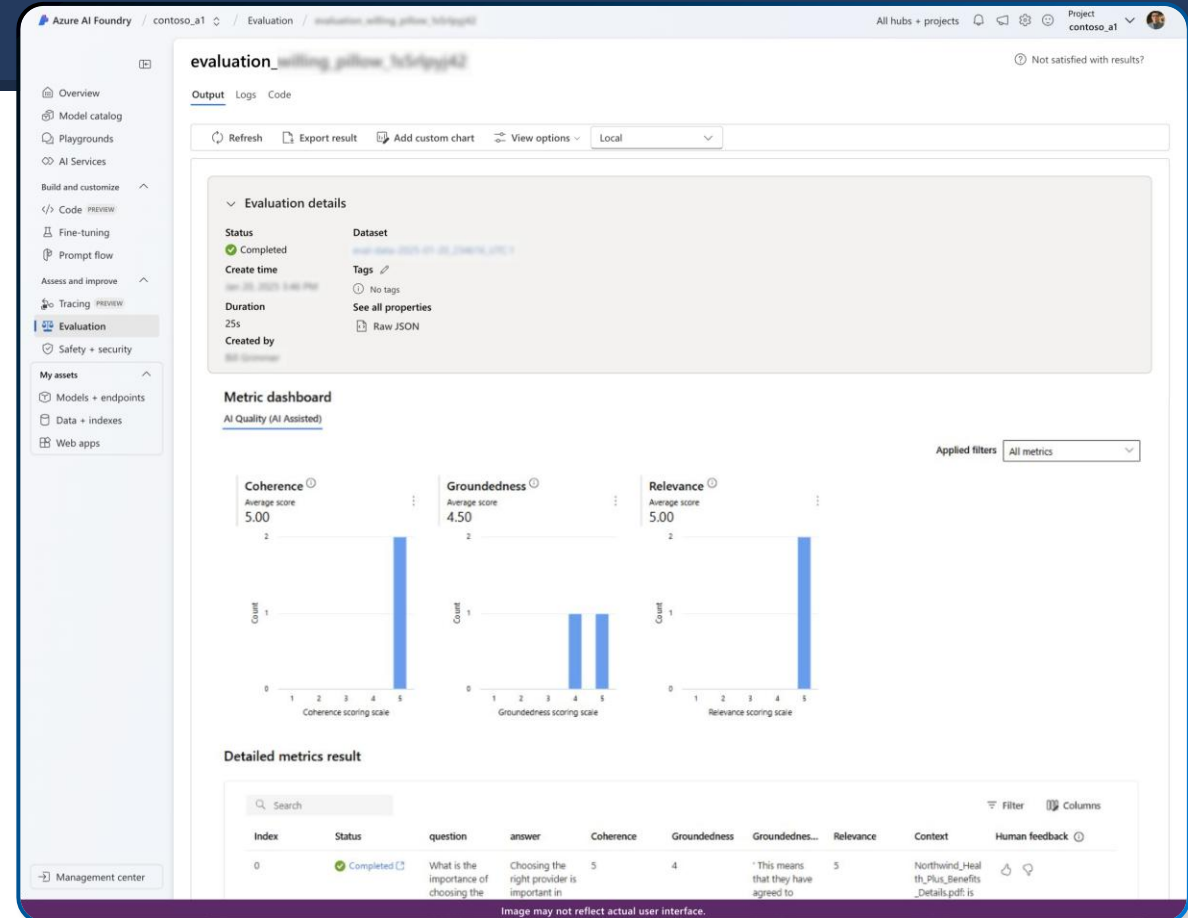
Custom

Customize pre-built
metrics or build your
own metrics or synthetic
data simulator with
Azure AI Evaluation SDK

Manual evaluation

Quickly and continuously iterate on your application and track the impact of ongoing changes by manually rating model outputs as you go.

- Manually create or upload your sample test dataset with the option to include expected outputs
- Customize your prompts to target specific aspects of model performance and user interaction you want to improve
- Provide a thumbs up or down rating to score each generated response
- Review model response scores by prompt or in the at-a-glance summaries
- Iterate on your application and re-run evaluations to track the impact of your changes
- Save and compare results to identify the optimal application design for your desired outcomes



Based on the results, you may decide to update your system message, model, or model parameters. Then, rerun the dataset or specific prompts that didn't meet your expectations to see the impact of your updates.

Evaluation datasets

Dataset	Description	Metrics/goals
Qualified answers	Small number of examples obtained from experts	Maximize response quality (groundedness, relevance, coherence, ...)
Synthetic	Large number of synthetic samples	Maximize response quality Maximize retrieval metrics (recall, precision) Maximize tool selection metrics (accuracy)
Adversarial	Jailbreak attempts, harmful content	Minimize unsafe responses
OOD	Out of domain questions	Minimize relevance ;-)
Thumbs down	Dataset of bad answers	Eye-ball, use to discuss
PROD	Scrubbed questions from opt-in users	Maximize response quality Ensure production satisfaction

AI innovation across landscapes



Visual Studio



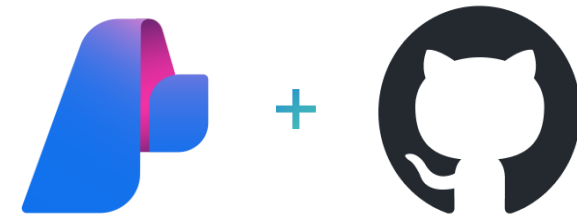
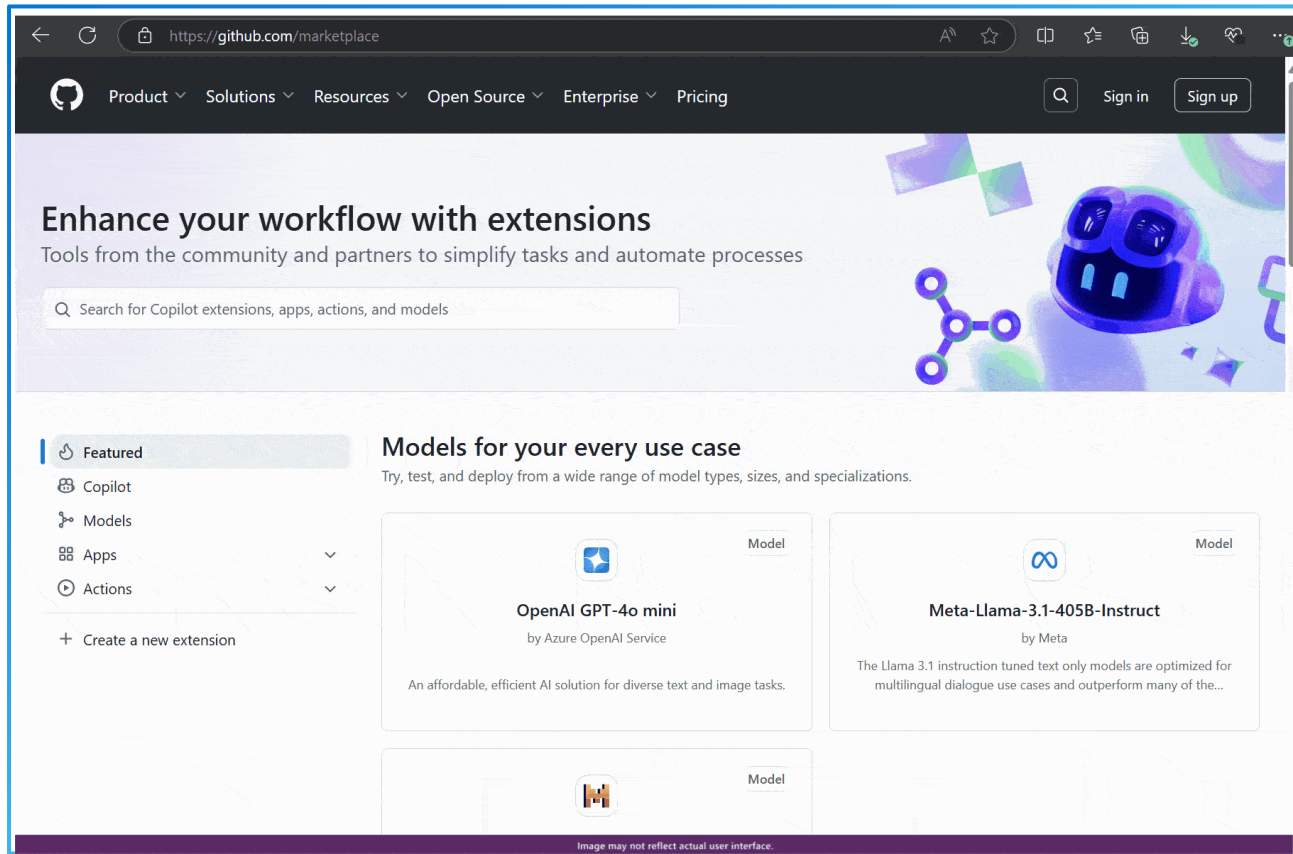
GitHub



Copilot Studio

World's **most loved** developer tools

Empowering GitHub Developers to Build with AI



- Get access to a curated list of our top AI models, without leaving GitHub
- Mix, match, and experiment with models in the GitHub playground for free*
- Unlock enterprise-level security and monitoring in production with Azure

Discover in GitHub Models
<https://github.com/marketplace/models>

Better together

Copilot Studio +
Visual Studio Code / GitHub,
all powered by Azure AI Foundry



Copilot Studio

Low-code managed
environment for AI business
process automation



Visual Studio / GitHub

Pro-code environment for
commercial AI solutions



Azure AI Foundry

Comprehensive platform of capabilities, models, and tools to design,
customize agents, and manage AI applications and agents

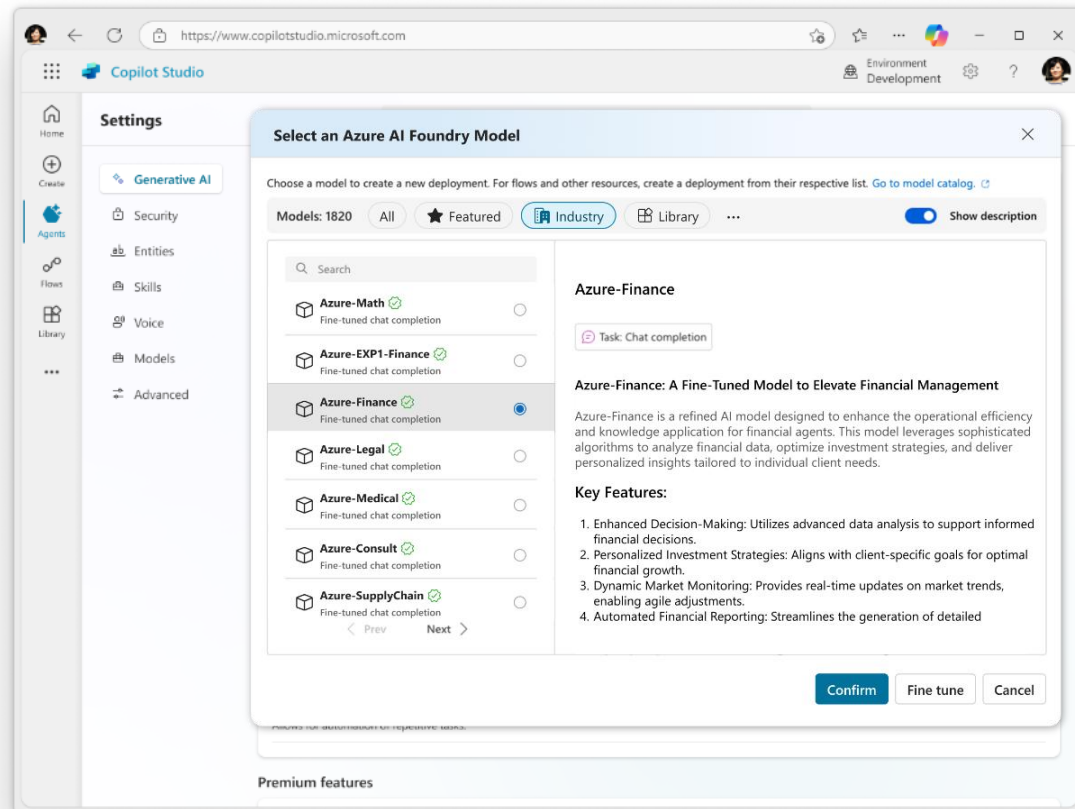


Microsoft Fabric

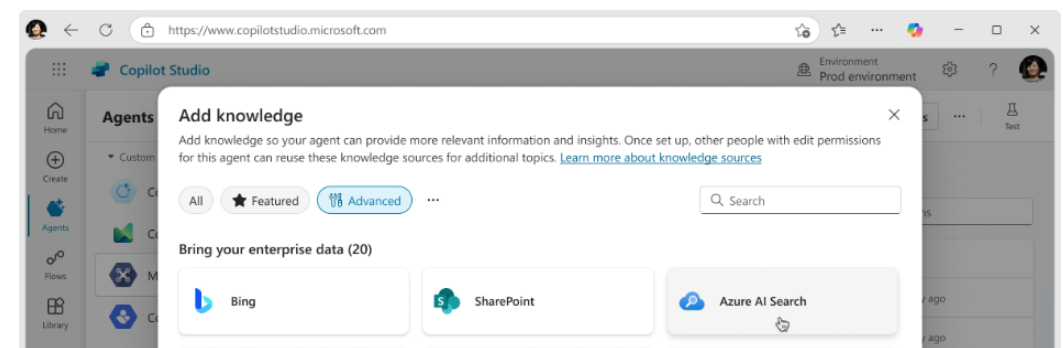
Complete data platform to accelerate AI innovation

Azure AI Foundry in Copilot Studio

Customers can discover and use models from Azure AI Foundry



Customers can access Azure AI Search as a knowledge source



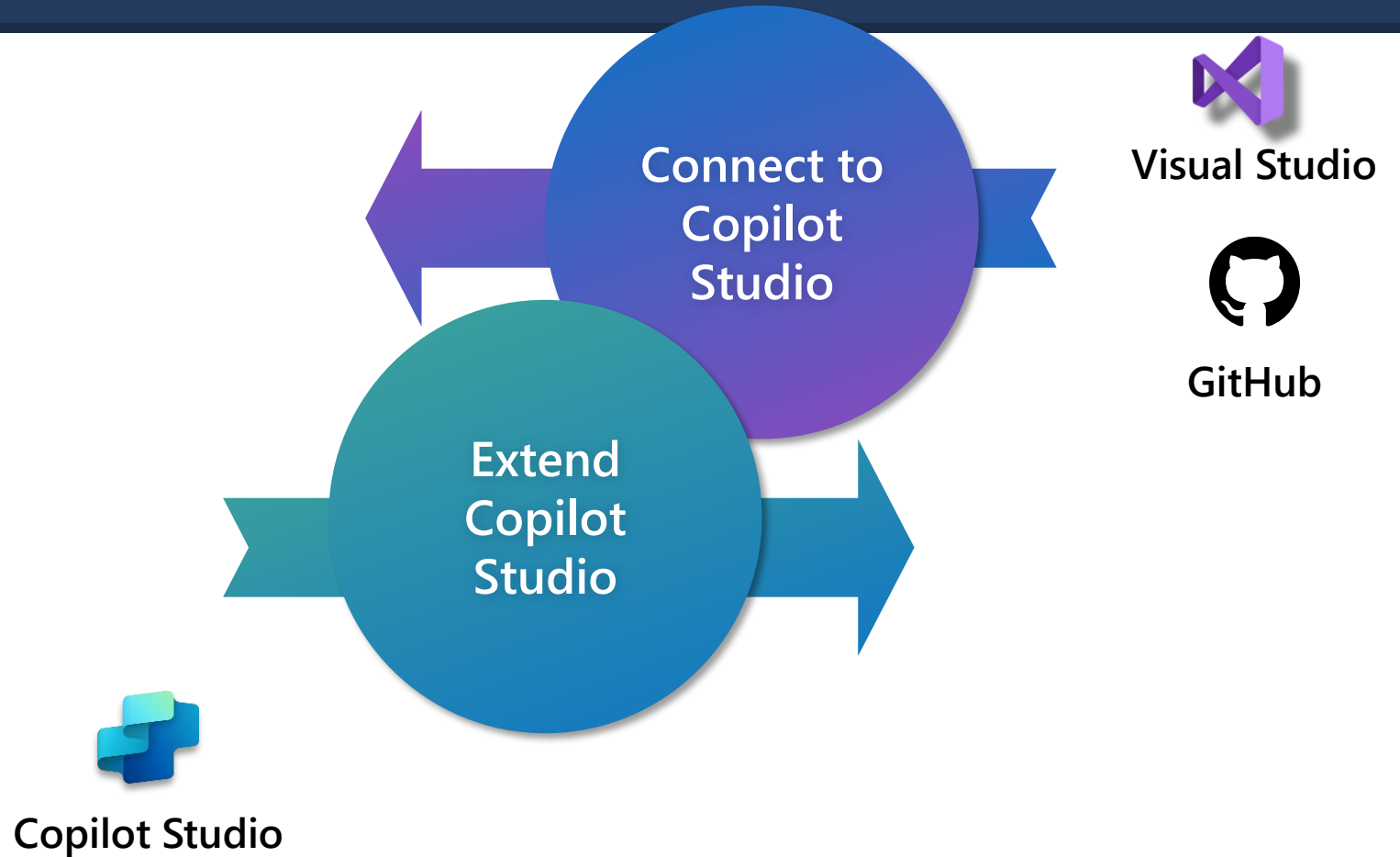
Microsoft 365 Agents SDK

Create agents with code, from conversational agents to multi-agent systems.

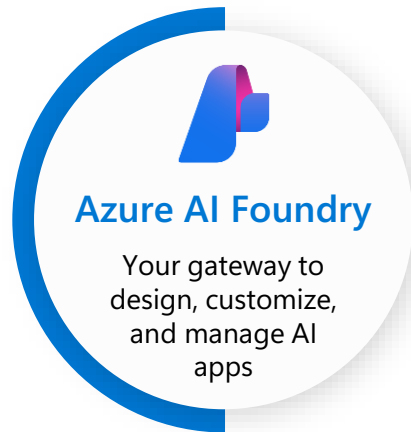
Leverage Azure AI Foundry, Semantic Kernel, and other AI services of your choice

Extend Copilot Studio with an agent, or use with agents built with Copilot Studio

Reach millions of users across Microsoft 365, web, messenger, and +10 channels.



Azure AI Foundry + Microsoft Fabric = boundless possibilities



Enhance data insights

Build enterprise chat applications to uncover insights using natural language from structured, unstructured, and real-time data

Ground your AI on unstructured data

Ensure your AI is using the latest data in its interactions with your employees, partners, and customers

Analyze customer interactions

Build speech analytics applications to enhance customer service, tailor support responses, and make data-driven decisions

Customize machine learning models

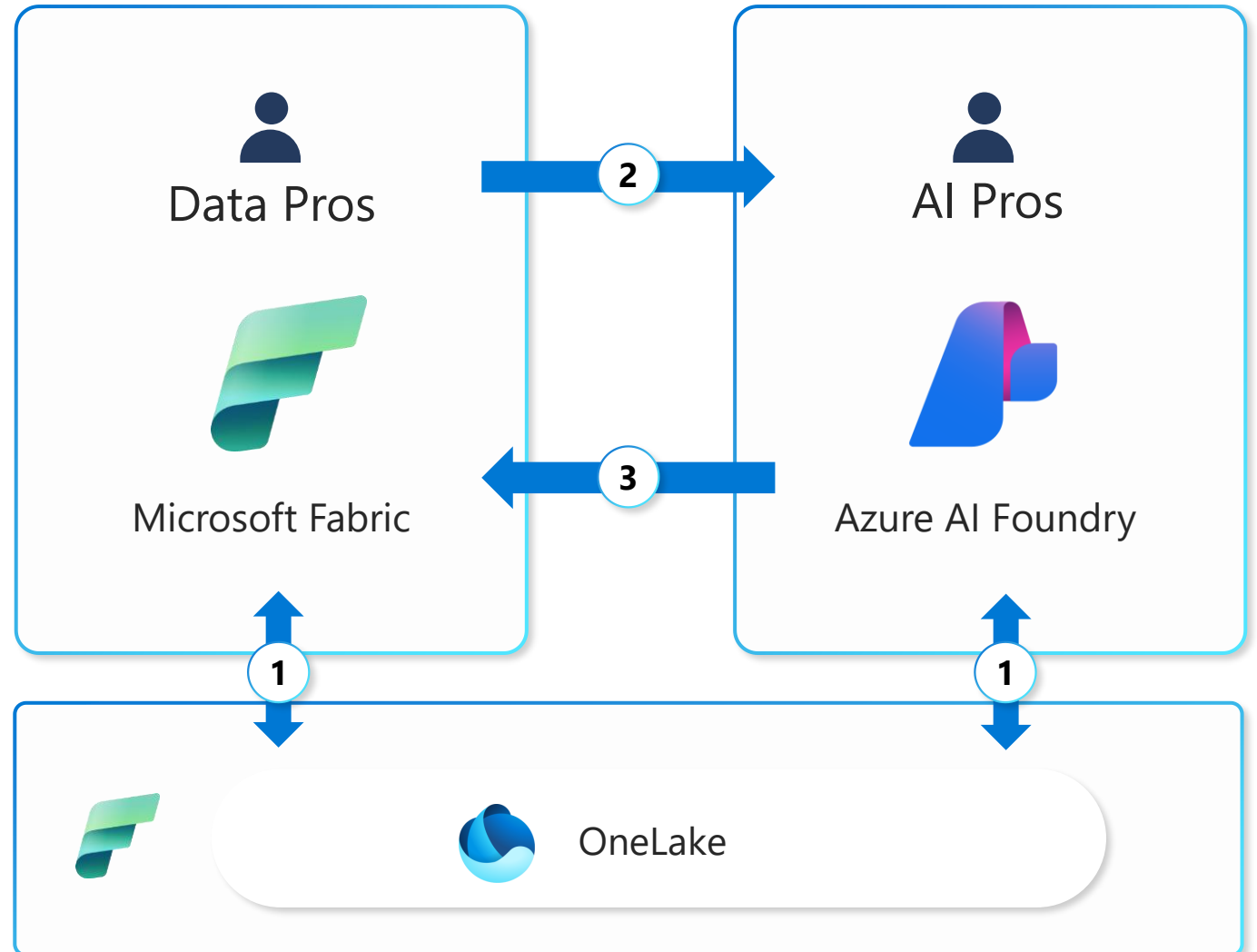
Train, deploy, and orchestrate machine learning models tailored to specific business needs—from predictive maintenance to customer sentiment analysis



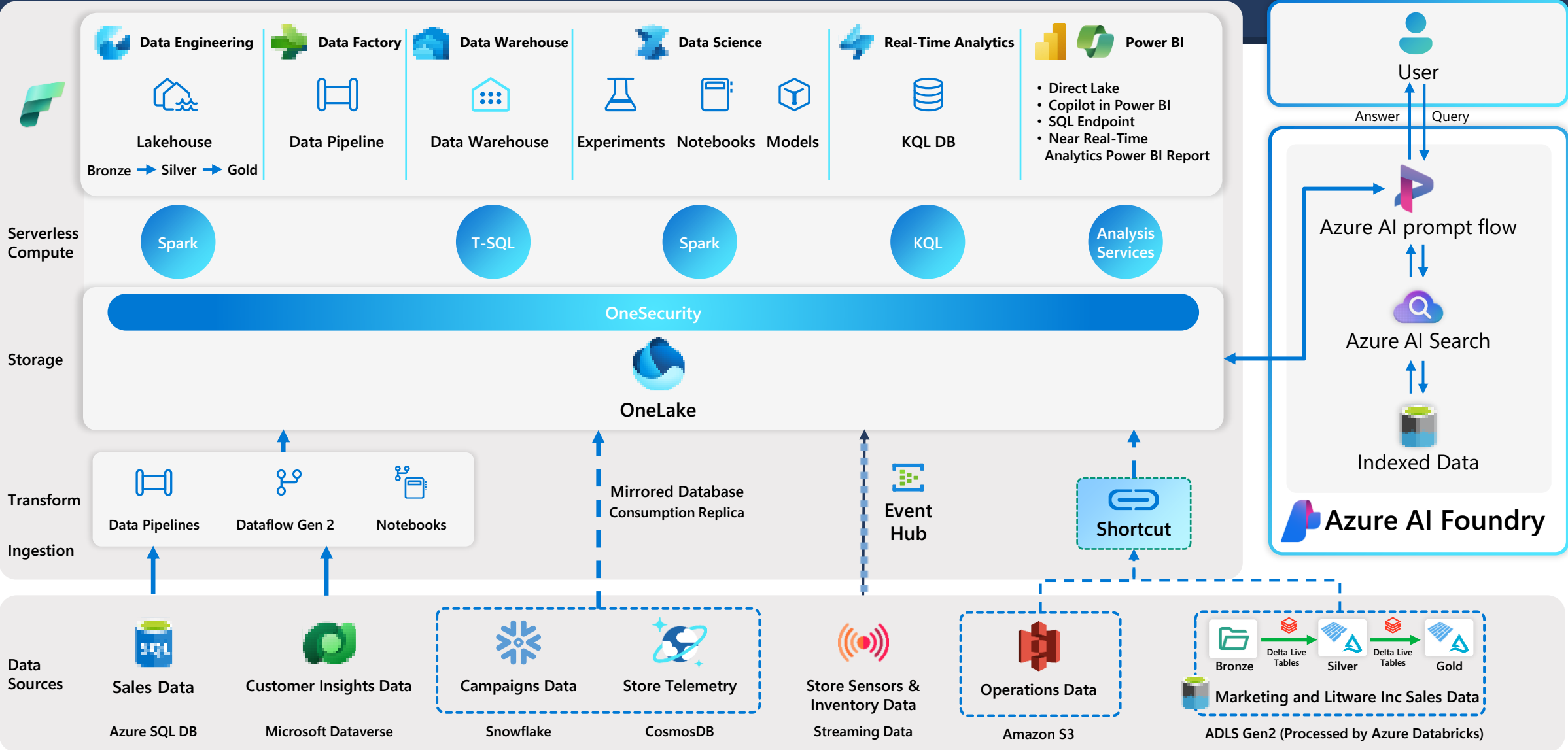
OneLake serves as the connective tissue to build generative AI apps powered by your data

Microsoft Fabric and Azure AI Foundry integration

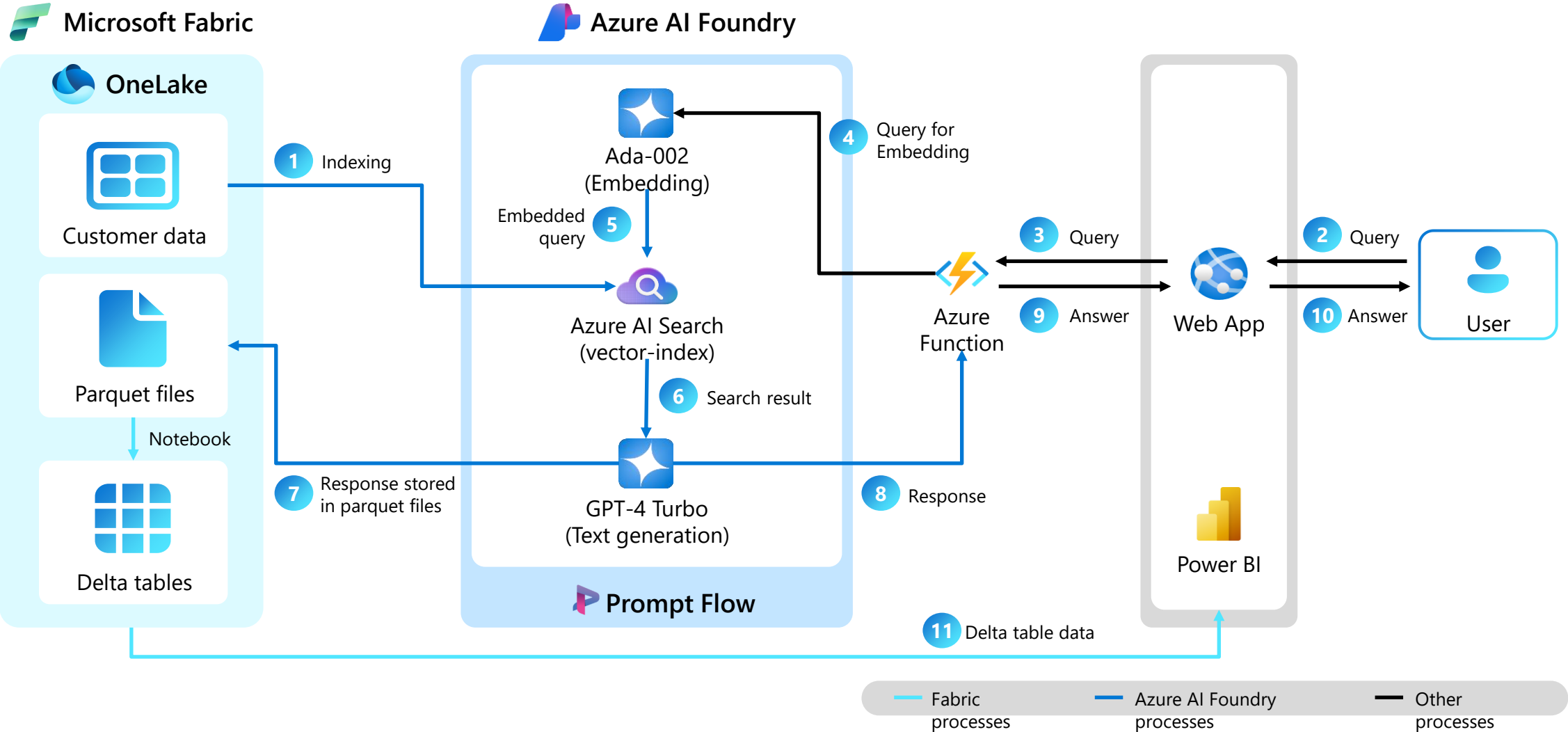
- 1** Data and AI Pros access, use and process data across the AI lifecycle with OneLake as the connective tissue.
- 2** Data Pros can pre-process and share AI-ready data with AI Pros. Once in Azure AI Foundry, AI Pros can create their own generative AI applications and copilot experiences.
- 3** AI Pros can make their models and generative AI apps accessible in Microsoft Fabric for Data Pros to enrich their analytics workflows in their lakehouse and serve through Power BI.



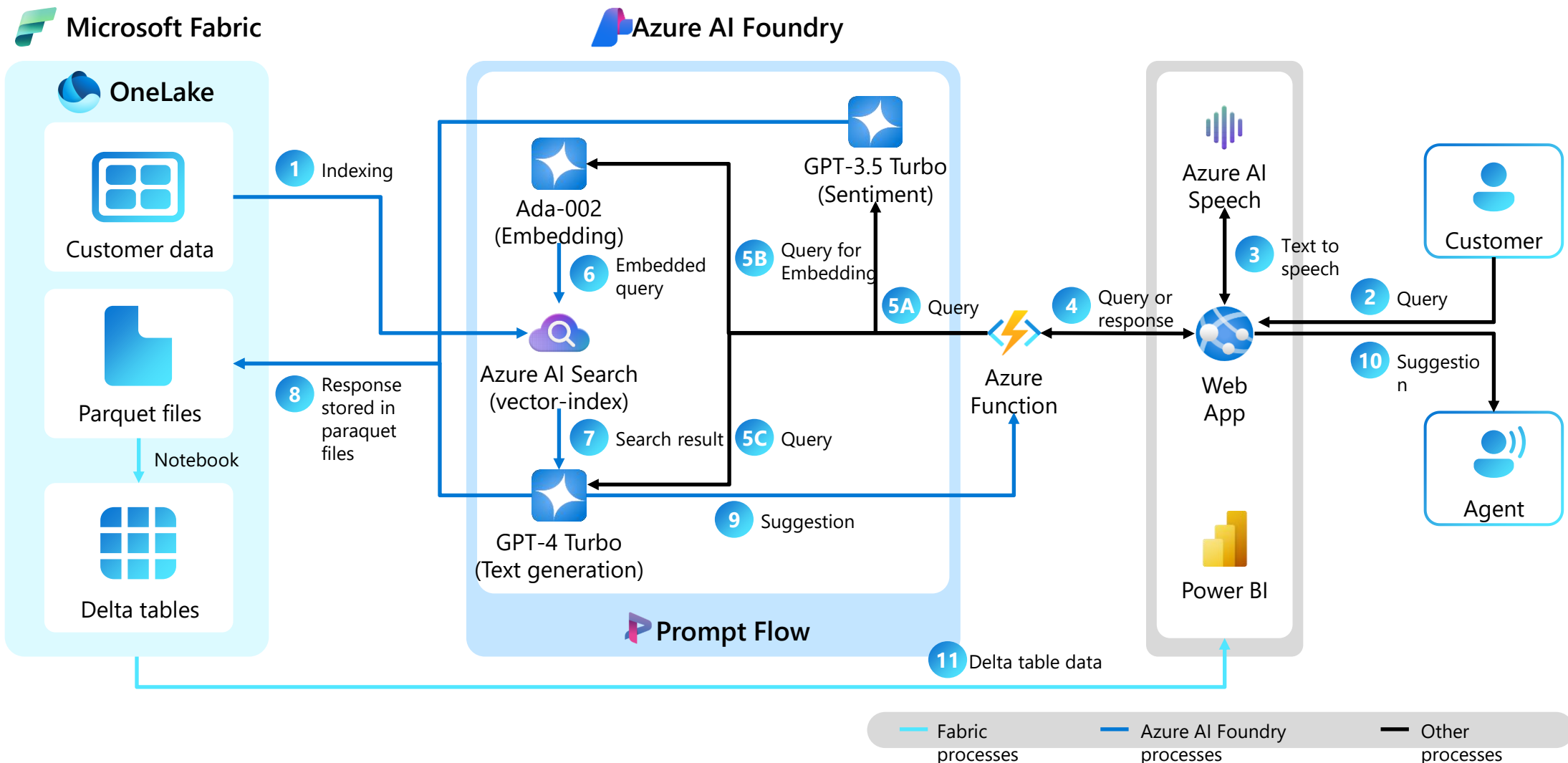
Microsoft Fabric and Azure AI Foundry architecture



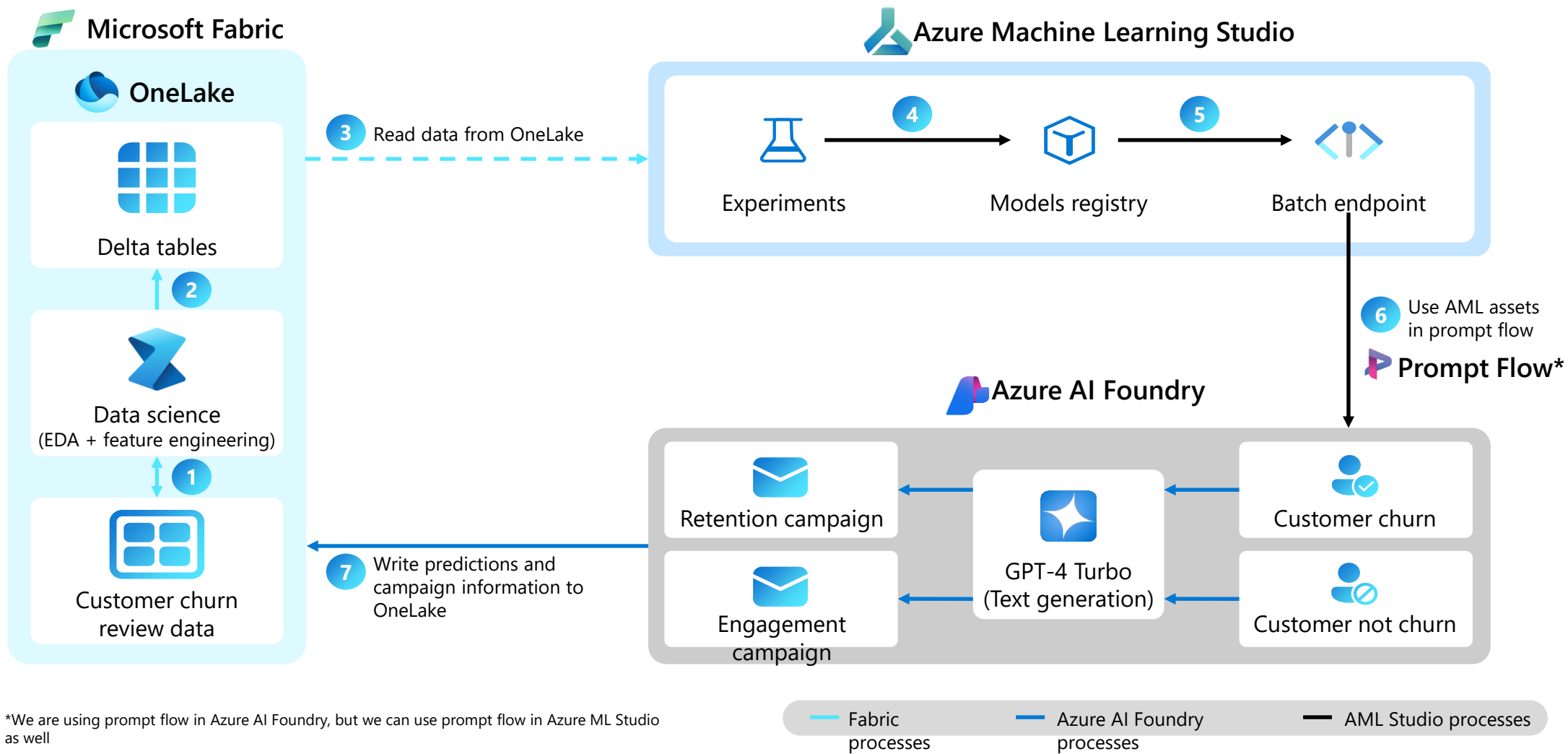
Enhance data insights



Analyze customer interactions



Customize machine learning models



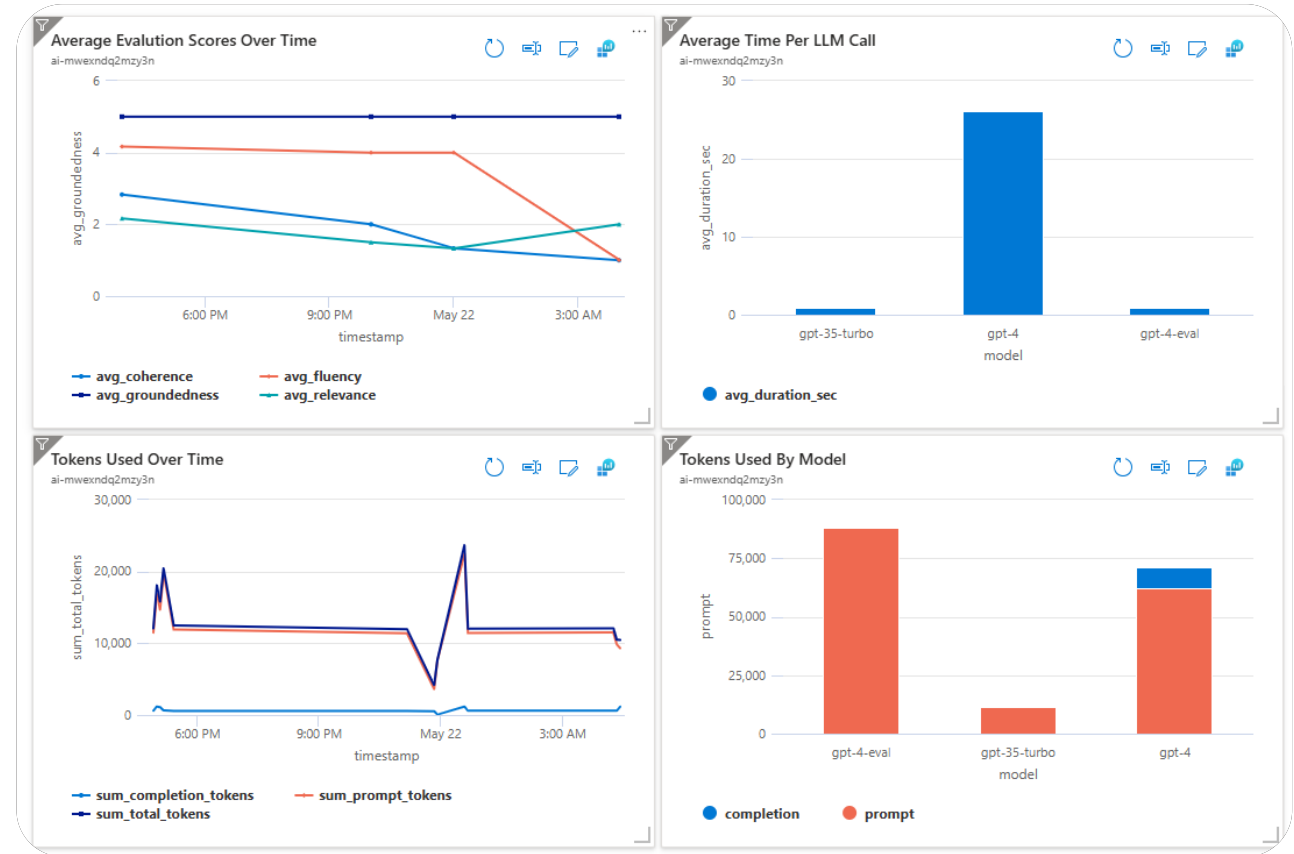
Monitor data in production with App Insights in Azure Monitor

Effortless production data logging

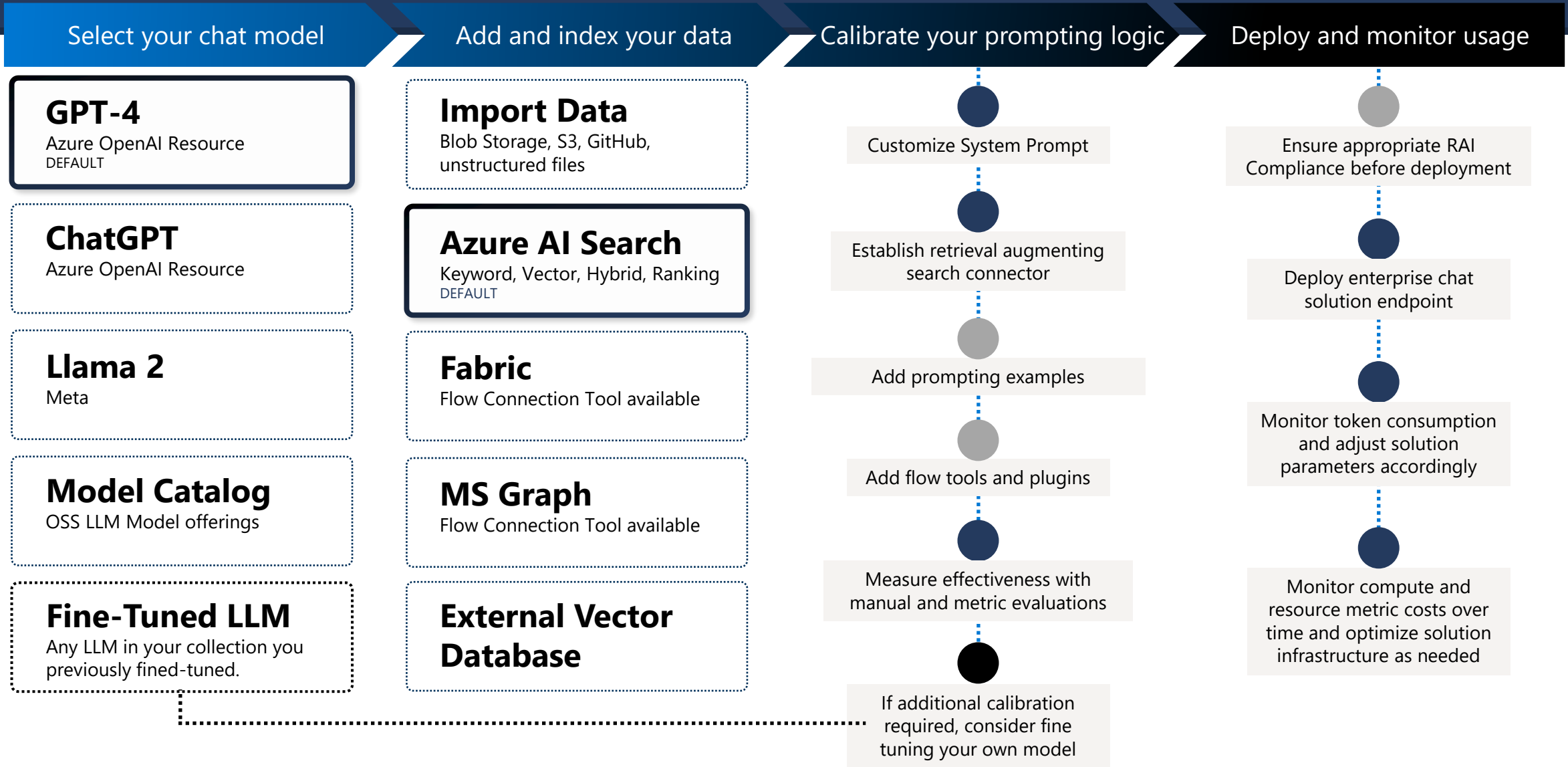
Enable monitoring for operational (error rate, latency) and token (token usage) metrics for GenAI deployments

View extensive monitoring results for your prompt flow deployment within a comprehensive UI

Configure alerts for violations based on organizational targets and run monitoring on a recurring basis



Chat with your data high-level architecture



Let's look at an example: Contoso Outdoor Company



Get the sample code on GitHub
[Contoso Chat](#)



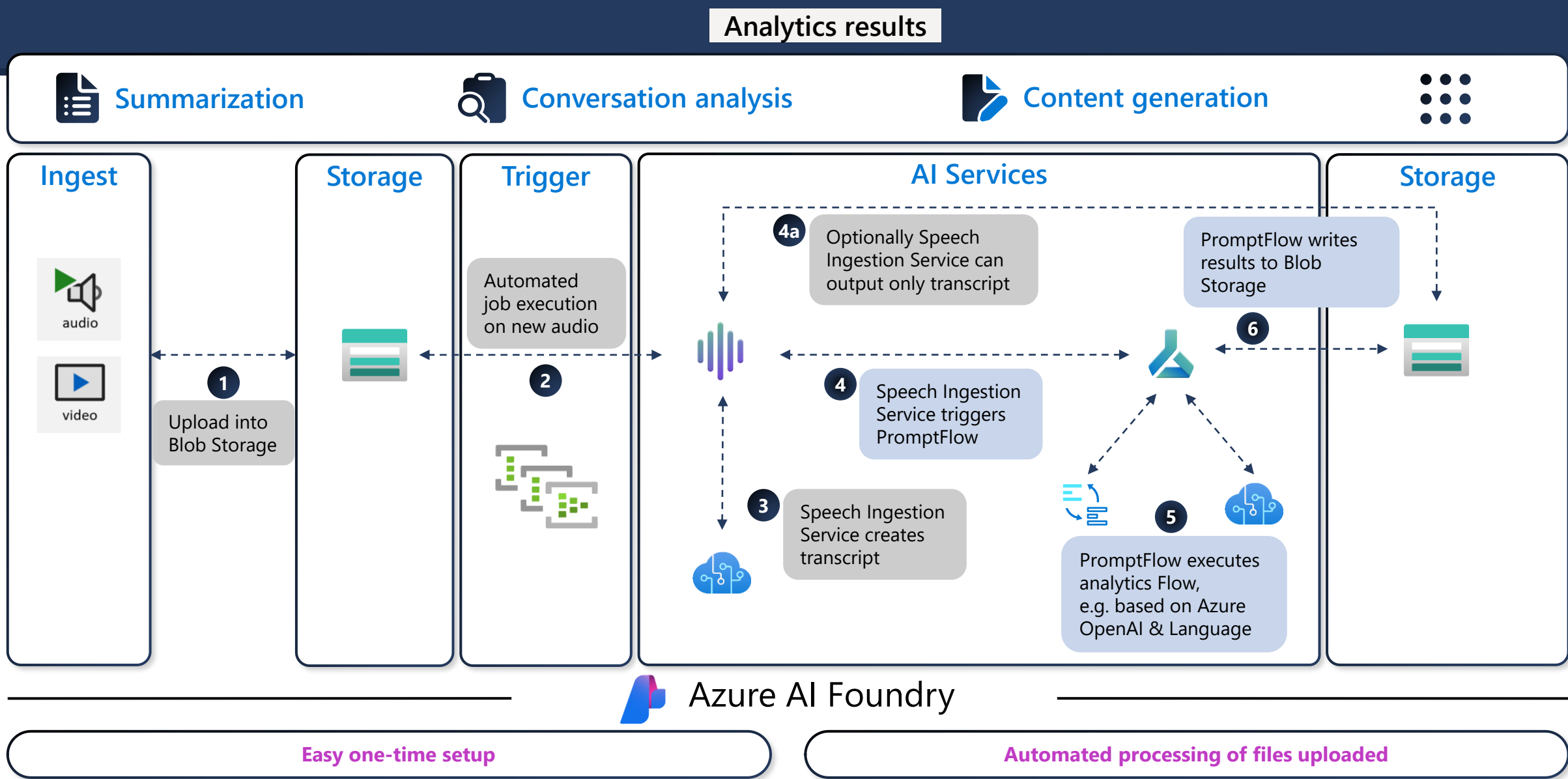
Images

Add your data

Azure AI Search + embeddings from Azure AI Vision

GPT-4 Turbo with Vision

Speech analytics high-level architecture



What's New In Azure AI Foundry - Models

o3-pro Enterprise Reasoning	o3-pro delivers the highest reasoning performance by using more compute to provide deeper, more consistent answers across complex enterprise tasks.
Sora Video Generation	Sora is now available in public preview in Azure AI Foundry, bringing enterprise-grade video generation capabilities. Create up to 20-second videos in 1080p resolution
codex-mini Lightweight Coding Assistant	The new codex-mini model joins Azure OpenAI as a lightweight, efficient option for code generation, completion, and programming assistance
Grok 3 & Grok 3 Mini Now GA	Grok 3 and Grok 3 Mini are now generally available in Azure AI Foundry
DeepSeek-R1-0528 Advanced Reasoning	DeepSeek-R1-0528 brings the latest evolution in reasoning-optimized models with improved depth of reasoning and inferencing capabilities .

What's New In Azure AI Foundry - Agents

Model Context Protocol Support	Bring any remote MCP server—self-hosted or SaaS—and Azure AI Foundry automatically imports capabilities, keeps them updated, and routes calls through enterprise security.
SharePoint Tool Integration	Seamlessly integrate agents with Microsoft SharePoint to chat with private documents securely.
Multi-Agent Workflows & Orchestration	Coordinate multiple specialized agents within unified workflows to handle complex, multi-step business processes.
Unified Semantic Kernel + AutoGen Runtime	single, composable API for defining, chaining, and deploying agents locally or in the cloud with identical behavior
DeepSeek-R1-0528 Advanced Reasoning	DeepSeek-R1-0528 brings the latest evolution in reasoning-optimized models with improved depth of reasoning and inferencing capabilities .
Agent Catalog & Code Samples	Access a centralized catalog of agent code samples with predefined instructions, actions, APIs, knowledge, and tools.

What's New In Azure AI Foundry - Tools

Enhanced VS Code Extension	Azure AI Foundry VS Code extension now supports full agent CRUD operations, YAML IntelliSense, and direct deployment
Trace Agents Debugging	New trace agents functionality provides detailed visibility into agent threads, displaying the exact sequence of inputs and outputs for each agent run.
Enhanced GitHub Development Platform	Manage Azure AI Foundry functionalities directly within GitHub repositories for seamless code collaboration and agent deployment.
Logic Apps Agent Triggers	Automatically invoke agents in response to events like new emails or customer tickets using Azure Logic Apps integration.
Agent-Specific Tools Expansion	New specialized tools include Bing Custom Search for grounding agents with web data and Morningstar integration for investment research

Lab - AI Foundry - Connection and First Steps with Azure OpenAI

Lab Description

- In this lab, you will take the first steps with Azure OpenAI Service, learning from initial configuration to advanced practical tasks. The lab covers API calls, detailed response analysis, embedding generation, image processing with multimodal models, and exploration of other LLM models available in Azure AI Foundry.

Lab - AI Foundry Observability and Evaluations

Lab Description

Observability

How to set up **Observability** and **Tracing** for:

- **Basic LLM calls** using an AIProjectClient.
- **Multi-step** interactions using an **Agent** (such as a Health Resource Agent).
- **Tracing** your local usage in **console** (stdout) or via an **OTLP endpoint** (like **Prompty** or **Aspire**).
- Sending those **traces** to **Azure Monitor** (Application Insights) so you can view them in **Azure AI Foundry**.

Evaluations

How to evaluate models with both **local** and **cloud** evaluation.